Software & Services

Unprofitable May Not Be Uninvestable; How Accounting Rules Dictate Profit Views

How to value an unprofitable subscription SaaS company? It is a question we get frequently today, especially against a rising interest rate backdrop where most industry pundits talk about how unprofitable companies are less desirable to invest in during the current Fed rate tightening cycle.

In this industry note, we discuss how profitability through a company's growth phase can be little more than the FASB accounting treatment of growth investments differing between sectors. A textbook example is WMT. While WMT was GAAP profitable at its IPO (year eight of its history), the company did not generate positive FCF until year 34 of its corporate life. Compare that to Workday (WDAY, Buy), which generated positive FCF in year 10 while not generating GAAP net income until year 16. The primary difference? Purely the capitalization of growth investments which sit on WMT's balance sheet vs. in period expensing for most of WDAY's investments.

This distinction is important today because strong, yet unprofitable, business models are akin to the old "baby and the bath water" slogan getting thrown out in the current macro chaos with companies that lack strong operating fundamentals. And why is this important? Because last time we checked in with our Business School Finance professor, **company valuation is the present value of all future cash flows, not the present value of future earnings,** which can be manipulated by accounting treatment at times. We believe the current macro has created attractive entry points for companies with these strong operating fundamentals, attractive unit economics and high gross margins.

The analysis in this note shows how the profitability of WMT changes if its growth investments are expensed in-period versus capitalized. This allows a more accurate comparison against a subscription business model like WDAY that already throws off better FCF margins (while expensing Share-Based Compensation) than a company like WMT ever will. Of course, the key is to identify which subscription software companies have the more attractive operating models at current valuations whose stocks can drive meaningful alpha over the intermediate-term. DOCS, PAYC, and VEEV already exceed the terminal value margin structure of our analysis and are well known to public software investors. Outside these, we highlight BILL, HUBS, ESMT, FIVN, PCTY, SMAR, and WDAY, each of whom demonstrates strong unit economics, a high revenue growth profile, and a long-term margin structure that creates a strong entry point versus its current valuation.

Topic of Discussion

Industry note to identify growth investments by sector and to highlight why an unprofitable SaaS company can be a positive investment idea when these accounting treatments are normalized between sectors.

CONTRIBUTING ANALYSTS

SCOTT BERG
(763) 350-4027
sberg@needhamco.com

RYAN MACDONALD
(763) 226-3120
rmacdonald@needhamco.com

JOSHUA REILLY, CFA
(612) 386-4640
jreilly@needhamco.com

JOHN GODIN
(612) 396-8402
jgodin@needhamco.com

MATTHEW SHEA
mshea@needhamco.com

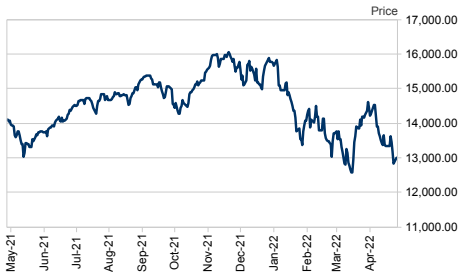
MICHAEL RACKERS
(314) 452-0719
mrackers@needhamco.com

MENTIONED COMPANIES

DOCS: $46.66, PT: $70.00
ESMT: $20.97, PT: $40.00
FIVN: $106.25, PT: $160.00
HUBS: $401.30, PT: $700.00
NFLX: $209.91, PT: NA
PAYC: $303.46, PT: $450.00
PCTY: $196.86, PT: $325.00
SMAR: $50.81, PT: $57.00

SECTOR CHART

Nasdaq (COMP)
04/26/22



RELEVANT DISCLOSURES BEGIN ON PAGE 34 OF THIS REPORT.

EXECUTIVE SUMMARY - CASH FLOWS DICTATE COMPANY VALUATION, NOT PROFITABILITY

Our business school finance professors taught us that company valuation is the present value of all future cash flows.

Take a moment and let that sink in.

And our accounting professors taught us that even profitable companies can go bankrupt.

While this bankruptcy scenario seems like a bit contradictory, let it also sink in. It can happen to a company if sales are sold on credit that is never collected or with certain accounting treatments, just ask the investors in Enron or Worldcom.

In this industry note, we compare the business model of the subscription software company against examples from eight of the 10 non-IT S&P 500 sectors to highlight why the current broad macro view of not investing in an unprofitable technology company is akin to the old saying of throwing out the baby with the bath water. Yes, some companies have less or unprofitable business models that should receive higher scrutiny in the current macro. However, our analysis drives a conclusion that profitability through a company's growth stage is a direct correlation to FASB's accounting standards for the treatment of growth investments and not necessarily due to a poor business model or a choice the company can make. We argue that if company valuation is defined as the present value of future cash flows, investors should focus on FCF and unit economics when comparing growth companies across industries, which really dictates the long-term profitability of most operating models, and not focus on earnings, which can make cross-sector comparisons imprecise.

In summary, we highlight that companies in most industries can show profitability through their respective growth phases because FASB rules allow for the capitalization of significant portions of these companies' capital-intensive growth investments versus FASB rules that force subscription software companies to expense the majority of growth investments up front. The result is a less profitable model in these growth years for subscription software companies but a more profitable long-term model that is without the burden of high D&A levels. We also note the subscription software operating model is tax efficient, due to high early GAAP net losses and Net Operating Loss carryforwards shielding future profits delaying the corporate tax liability associated profitability.

We believe a comparison of Walmart versus Workday (WDAY, rated Buy) accentuates our view point. Walmart was GAAP profitable effectively at its IPO but did not generate positive FCF until its 34th year. However, Workday generated positive FCF in its 10th year but did not generate GAAP profitability until year 18 or Non-GAAP profitability until year 13. The primary difference is purely accounting driven. The majority of Walmart's growth investments are capitalized versus Workday's growth investments, which are expensed as incurred. A proper comparison of business models would be to either capitalize the Workday investments or immediately expense the Walmart investments (which we do). If one expenses the Walmart investments as incurred, the company looks significantly unprofitable for decades, yet the model as reported shows a strongly profitable business.

In this note, we also highlight our thought process on how to value high growth SaaS company through DCF-based modeling. While we believe this theory based thought process supports median EV/Forward Year Revenue valuations of 10x-12x due to revenue growth rates that are higher and more durable than expectations just five or 10 years ago, we are mindful that this viewpoint requires strong unit economics and a high level of subscription gross margin to drive appropriate profitability and cash flows as the investments slow. This framework does place a higher weighting on revenue growth in the model over near-term profitability given the typical growth-compounding effect that drives much higher cash flow generation over time. Through this lens, we identify several companies in our coverage universe with attractive business models that we believe are under-appreciated at current valuation levels. DOCS, PAYC, and VEEV have attractive business models that are well known to public software investors given their current FCF generation levels against strong revenue growth rates. The other companies with excellent unit economics and gross margins that we believe can drive operating models above our 30% terminal operating margin structure are BILL, HUBS, ESMT, FIVN, PCTY, SMAR, and WDAY,

COMPARING ACROSS SECTORS: ACCOUNTING TREATMENT OF GROWTH INVESTMENTS IS KEY DIFFERENCE OF TIMING TO PROFITABILITY

Our journey to review and compare business models across sectors quickly took us to analyzing the differences between how revenue and expenses are recognized. Most investors already understand the differences between revenue recognition for a subscription recognized ratably over a contract's life versus revenue that is (generally) recognized at the point of sale, the model for essentially everything else. We acknowledge that this revenue recognition timing for non-subscription software companies can vary at times based on the delivery of

product versus the date of sale, but the revenue tied to a widget's sale or service tends to be all at once in the majority of circumstances. We summarize that the ratable recognition of subscription revenues is a contributing factor to delaying profitability, but note these revenues, especially on a contract's renewal, tend to turn into a profitability tailwind due to the lack of direct acquisition costs versus an initial sale.

Our analysis on expenses identified the largest difference across the sector's business models: treatment of growth investments. In the eight non-IT sectors highlighted in Figure #1 below, every company capitalizes some, if not most, of its respective growth investments. However, with subscription software, the majority of growth investments are people within R&D or sales teams, which FASB requires the companies to expense as the costs are incurred in most cases. We view the sectors' growth investments through the following lenses:

- **Subscription software:** Workday's primary growth investments are in R&D and S&M. The majority of these expenses are recognized as incurred. We note the changes for amortizing sales commissions within ASC 606 now capitalizes some of this expense. We also note that Workday does capitalize some owned real estate costs, but this practice is unique within most subscription companies and more reflective of its massive size and decision to own office space versus leasing.

- **Consumer Staples or Consumer Discretionary:** Walmart's growth investments are primarily property and buildings. Home Depot is almost identical to Walmart while Chipotle as a restaurant also has similar CapEx outlays.

- **Healthcare:** Healthcare broadly capitalizes growth investments due to high levels of CapEx for product development and facilities, yet the pharmaceuticals looks a lot like a SaaS model with heavy levels of R&D expensed as incurred.

- **Communication Services:** Communication services was the trickiest sector to analyze given a wide range of business models in the sector. However, most companies also capitalize high amounts of growth investments. AT&T and Charter of course has buildings and infrastructure to capitalize. Netflix (NFLX, rated Hold by Laura Martin) at first looked at though it would be similar to a subscription software company, but Netflix is able to capitalize much of its content costs, which improves profitability early in its life.

- **Industrials, Materials, Energy, and Utilities:** Unsurprisingly, most companies in these four sectors capitalize heavy CapEx/PP&E expenditures.

Figure 1 - Time to GAAP Profitability and Positive FCF by Company/Sector

Company	Ticker	S&P500 Sector	Year Founded	Years to IPO (from founding)	Years to GAAP Profitable (from founding)	Years to FCF Positive (from founding)
Workday	**WDAY**	**Information Technology**	**2005**	**7**	**16**	**10**
Walmart	WMT	Consumer Staples	1962	8	1*	34
Home Depot	HD	Consumer Disc.	1978	3	2	23
Chipotle	CMG	Consumer Disc.	1993	13	11	13
Netflix	NFLX	Communication Services	1997	5	6	N/A
Old Dominion Freight Line	ODFL	Industrials	1934	57	1*	79
Dexcom	DXCM	Healthcare	1999	6	20	15
Vertex Pharma.	VRTX	Healthcare	1989	2	28	27
LyondellBasel	LYB	Materials	2007	3	3	3
Diamondback Energy	FANG	Energy	2007	5	10	13
NRG Energy	NRG	Utilities	1989	11	15*	26

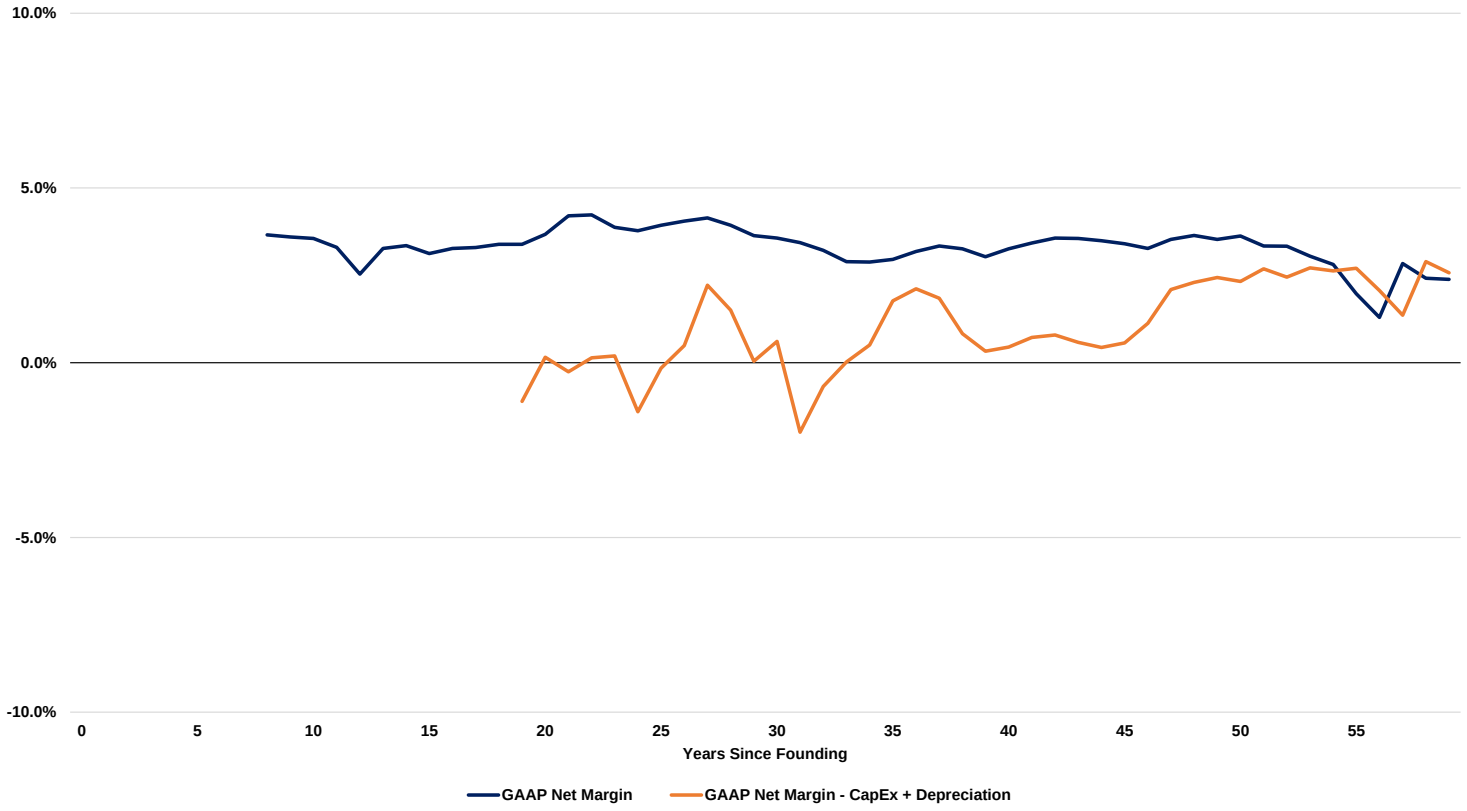
Source: Factset, Needham & Company Research

IMPACT OF EXPENSING GROWTH INVESTMENTS: LESS PROFITABLE MODEL EARLY, VASTLY MORE TAX EFFICIENT

As we further evaluate the differences in this accounting treatment for growth investments, the impact is readily apparent in that costs expensed as incurred reduce the profitability of a company's operating model through its growth stages. To illustrate this impact, we compare Walmart's annual GAAP Net Income vs what Net Income would look like if its CapEx were expensed as incurred. We acknowledge a company like Walmart does have other "investments" on the balance sheet it pays for upfront like inventory (although today it has a negative cash cycle on inventory) but these other impacts to cash flows/expenses are relatively immaterial compared to its early CapEx investments. To calculate this adjustment, we subtracted annual CapEx expense from GAAP Net Income but add back any depreciation to adjust out the impact of amortized CapEx from prior periods in the reported GAAP Net Income. As Figure #2 below highlights, Walmart's

Net Income would have been four or more percentage points lower for its roughly first 30 years including the first 20 that would be been unprofitable under this definition.

Figure 2 - Walmart GAAP Net Income vs Net Income less CapEx plus Depreciation Addback



Source: Needham & Company LLC, Factset, Company materials

A second, and less-appreciated impact, is how much tax efficient the subscription business model is based on FASB accounting rules. Expensing growth investments as incurred drives a less profitable or unprofitable model through a subscription software company's growth phase. Often, this lower profitability accumulates a relatively large Net Operating Loss Carryforward. The result is a business model that pays less or no taxes early in its growth phase and then shields a portion of its subsequent profitability with accumulated NOLs for a period of time that is a function of the level of profitability and the size of the NOLs. Although the nominal amount of taxes paid (in a static tax rate environment) would be equal over time since this exercise is just shifting the timing of expense recognition, the Time Value of Money theory suggests the present value of these tax liabilities can be significantly less given the accounting treatment for subscription business models.

VALUING A NO GROWTH SUBSCRIPTION SOFTWARE COMPANY THROUGH A DCF LENS

In this section, we take a look at establishing a baseline format to value subscription software companies. Our baseline centers on the traditional DCF valuation framework, which we simplify by growing a hypothetical subscription software for a set number of years and then assuming zero revenue growth whose value is relatively straight forward to calculate using a dividend discount model.

We set a control valuation for this exercise using a simple, no growth SaaS company with the following characteristics and rationales. We use this control scenario because it is akin to analyzing a mature, low growth company in almost any sector.

- **Annual revenue compression of -2.5%.** Our control scenario assumes no net sales either through new customer acquisition or upsell/cross-sell motions. This no-growth model assumes 5% annual dollar attrition, yet annual pricing increased at the historical rate of inflation at 2.5% to net out at 2.5% annual revenue compression. We acknowledge one could argue for a higher attrition rate as most enterprise-focused vendors have attrition closer to 7% - 8%, but we could also argue for higher annual rate of pricing increases

given we see many mature vendors today increasing effective annual pricing by 5% - 7%, which can be lumped into a 15% - 20% price increase for a new 3-year contract, or close to a 5% - 7% annual increase.

- **Peak Operating Margins are 30%.** We believe most mature SaaS companies can easily generate Non-GAAP operating margins of 30% or better in a no growth phase since growth investments are essentially non-existent. As we break down all expenses below, we could argue for Non-GAAP operating margins closer to 40% with COGS and all operating expenses really only at 60% of revenues, but we feel a 30% level is conservative enough to account for the cash compensation or share-based dilution from Share Based Compensation expense.
  - **Gross Margins:** Non-GAAP gross margins of 80% or better in a no growth stage which makes data center capacity and support requirements extremely predictable. Given we see most SaaS companies having subscription gross margins in the mid-70% to low 80% range today, this gross margin level should be achievable especially with minimal low margin services revenues with no net new sales.
  - **R&D Expense** We assume R&D expenses should run at roughly 15% of revenues to maintain a modern application so customer attrition does not increase at a faster rate.
  - **G&A Expense:** We assume G&A runs in the 7% - 8% range.
  - **S&M Expense.** While S&M expense in a no growth mode could equal zero, we assume 15% of revenues is required to maintain historical attrition levels.

- **Fully taxed at 21% Federal tax rate.**

- **Discount rate of 8%.** Since the majority of SaaS companies are financed with equity, we believe a 5% equity risk premium remains consistent with historical trends. Given the 40-year trend of lower 10 year (i.e. risk free) rates, and the last 20 years effectively averaging about 3%, we believe an 8% discount rate to be appropriate today and should remain durable for the foreseeable future.

- **Working capital adjustments equal zero and CapEx equals depreciation expense.** With a no growth company, we would expect the company requires no real working capital as well as CapEx requirements should equal annual depreciation. Given the slowing declining revenue levels in this scenario, one could argue negative working capital and CapEx requirements but for this exercise, we retain our conservative view and do not include any benefit.

Using a Dividend Discount model, this no-growth SaaS company would be valued at 2.26x current year revenues.

Figure 3 - Valuing a No Growth SaaS Company Using a DDM

Revenue		$1
Operating Income	$	0.30
Operating Margin		*30%*
Tax (@ 21% Fed Rate)		0.063
Net Income	$	0.237

$$\text{DDM} = \frac{\text{Dividend}}{\text{WACC} - g} = \frac{0.237}{0.08-(-0.025)} = 2.26$$

Source: Needham & Company Research

VALUING GROWTH SUBSCRIPTION SOFTWARE COMPANY THROUGH A SIMILAR DCF LENS

To value a growing subscription software company, we use a similar DCF-based lens but assume some level of sustainable growth for a set timeframe. One the set growth period ends, we assume the company's revenue growth rate immediately goes to zero and value the future period revenues using the same exact DDM scenario as defined in the prior section, of course discount back to the current period. The

sensitivity analysis in the following chart illustrates the growth scenarios with their respective EV/Revenue multiples. Our assumptions in this chart are fairly straight forward and are characterized by:

- **Years of Growth Stage:** The number of years a subscription software company will grow with annual growth in the subsequent years to be zero. Note though as in our example above, revenues actually compress 2.5% annually due to attrition partially offset by annual pricing increases.

- **Margins/Cash Flows:** We assume a Rule of 40 can be attained for each growth level. For the 0% and 10% and terminal margins, we assume 10% operating margins. For 25% annual revenue growth, we assume 15% operating margins. For 40% annual revenue growth, we assume 0% operating margins. For 50% annual growth, we assume -10% operating margins.

Figure 4 - Valuing Growth SaaS Company Using DCF

Growth Rate	Years of Growth Stage			
	4	8	12	16
0.0%	2.40x	2.55x	2.66x	2.74x
10.0%	3.36x	4.61x	5.95x	7.40x
25.0%	4.64x	9.02x	16.88x	30.99x
40.0%	6.21x	17.55x	49.55x	139.90x
50.0%	7.42x	26.85x	99.13x	368.12x

Source: Needham & Company Research

For example, a company growing 10% annual for four years, with growth immediately declining to 0% in year five, with a 30% operating margin structure has cash flows that support a 3.36x EV/Revenue multiple. If that company can continue growing at a 10% annual rate for eight years instead of four, with year nine immediately declining to 0%, the multiple expands 1.28x to 4.64x.

However, we have yet to see any subscription software company whose revenue growth rate immediately declines to zero. Rather, annual revenue growth can be sustainable for a multi-year period before declining to something other than zero. To value this scenario in Figure #5, we look at a company growing revenues at a certain annual rate and then reduce that revenue growth rate by 50% for the subsequent four years before than dropping to zero the year after this incremental four-year period. On the margin side, we use a similar Rule of 40%. As revenue growth is cut in half, margins instantly expand to drive a Rule of 40 metric.

Figure 5 - Valuing Growth SaaS Company Using DCF

Growth Rate	Years of Growth Stage	
	4	8
10.0%	3.36x	4.06x
25.0%	4.64x	7.07x
40.0%	6.21x	11.81x
50.0%	7.42x	16.50x

Source: Needham & Company Research

To view the chart above, for example, a 25% annual revenue growth company whose growth is reduced to zero can be valued at 4.64x revenues just as in Figure #5 above. However, with additional growth in years five through eight of 12.5% and operating margins expanding from 15% at the 25% growth rate to 27.5% in years five through eight with 12.5% revenue growth, the valuation multiple expands from 4.64x to 7.07x. For a 40% growth company with zero operating margins its first four years then has revenue growth cut to 20% in years five through eight with 20% operating margins, the valuation multiple increases to 11.81x. We note that both of these scenarios yield valuation multiples that are roughly just less than 50% of the increase than if the companies were able to sustain the original revenue growth rate for the entire eight-year period which we believe seems reasonable given the revenue growth rate was cut in half in the exercise.

Lastly, we model the same growth exercise for a company growing revenues at a consisted rate for eight years than the growth rate is reduced by 50% for years nine through twelve. We assume the same impact on margins rising to attain a Rule of 40 level. We then assume revenue growth immediately drops to zero in year 13 and has 30% operating margins. This scenario yields a similar result where the multiple expansion achieved from growing at half the revenue growth rate for four additional years yields a valuation multiple roughly in the middle than if the company grew at its original rate for the subsequent four years.

Figure 6 - Valuing Growth SaaS Company Using DCF

Growth Rate	Years of Growth Stage	
	8	12
10.0%	4.61x	4.87x
25.0%	9.02x	13.38x
40.0%	17.55x	33.35x
50.0%	26.85x	60.64x

Source: Needham & Company Research

We believe the net takeaway from this exercise shows that Finance theory supports a view that valuation multiples are more sensitive to revenue growth rates than profitability as long as operating margins at maturity fit the expected 30% or higher levels.

CONSUMER STAPLES - COMPARING WDAY TO WMT

Our first subscription software to S&P 500 sector comparison is with Walmart in Consumer Staples. We use Walmart due to its maturity and significant years of publicly available financial reporting data. We use Workday throughout this exercise because of its maturity level and its significant growth both before and after its IPO signifies that amount of early investments are often required in the subscription software model to support fast growth. We detail several components of the two companies' respective profiles below but summarize each for this exercise:

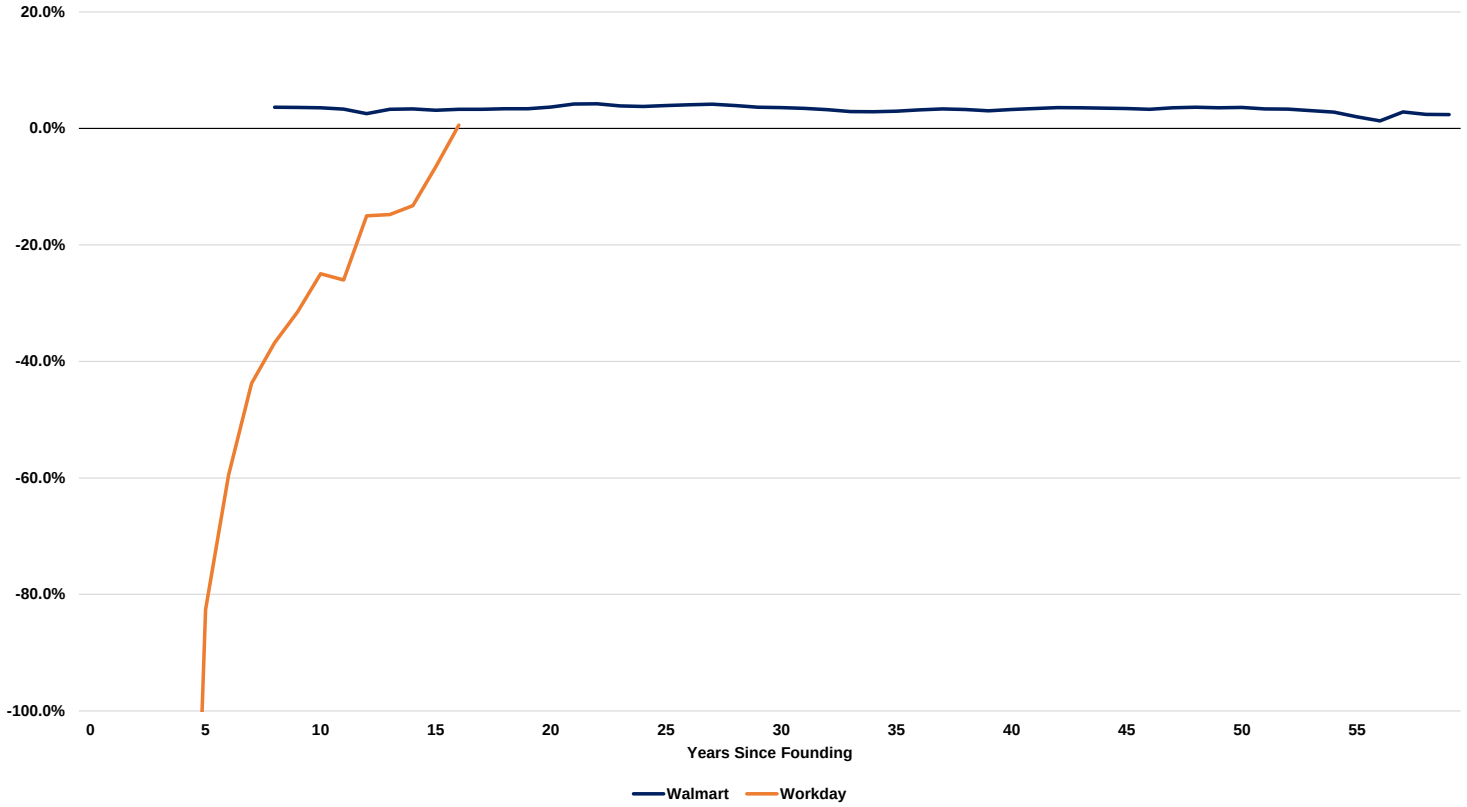
Workday (WDAY, Buy) generated its first annual positive FCF 10 years post founding, but no positive GAAP Net Income until year 16. Workday was founded in 2005 and went public seven years later in 2012 (FY13). While the company did not generate positive GAAP EPS until 2021 (FY22) due to heavy S&M and R&D investments throughout its growth phase, it generated positive FCF annually for the first time in 2015 (FY16).

Walmart (WMT, NR) generated its first annual positive FCF 36 years post founding, even though it showed positive GAAP Net Income in year 8. Walmart was founded in 1962, came public eight years later in 1970 and was listed on the NYSE in 1972. Walmart has been GAAP profitable virtually every year since its IPO, however it took nearly 30 years as a public company to turn free cash flow consistently positive. Free cash flow margins have been capped over the decades, never surpassing 5%.

WDAY vs WMT Comparison

In figure #7 below, we first compare the GAAP Net Income (or profitability) of each company. As the figure highlights, Walmart has been GAAP profitable every year since before its IPO while Workday just crossed this threshold for the first time in Fiscal 2022. Note the X Axis in the chart years since each respective companies' founding.

Figure 7 - GAAP Net Margins



Source: Needham & Company LLC, Factset, Company materials

Walmart's GAAP net income margin has been consistently around 5% since public numbers are available starting in 1970. Workday net margins tell a different story, taking until 2021 to turn positive after years of accelerated improvement from a very low level. We view this leverage observation as important to understanding the difference in growth investments between the two companies. The majority of Walmart's growth investments appear not on the income statement, but on the balance sheet. When a newly constructed store starts operating, its GAAP Net Margin quickly climbs to the corporate average. On the Workday side, any growth investments are initially margin dilutive with the company's gaining the leverage through higher sales productivity or a larger customer set paying for the newly developed functionality.

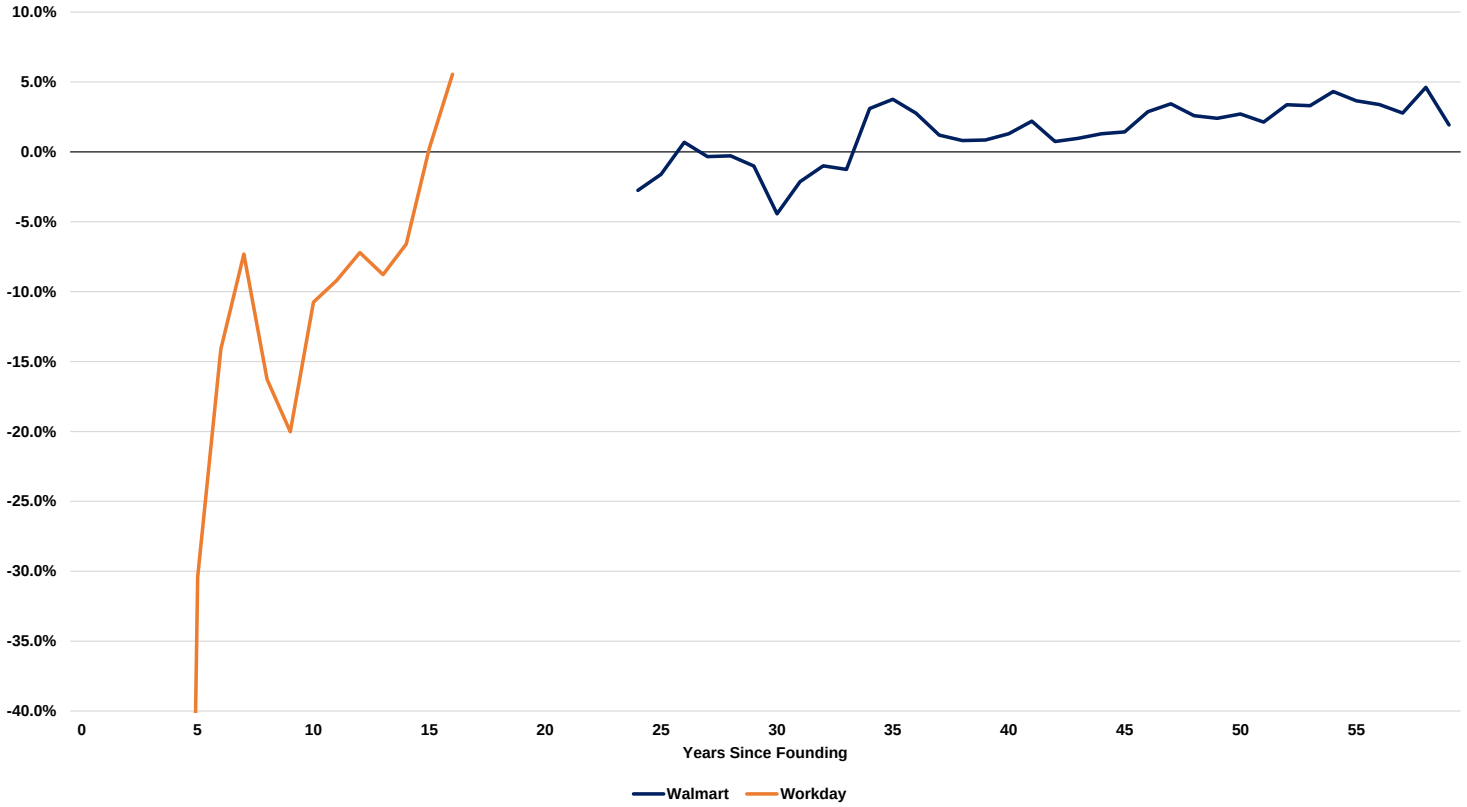
However, if we look at FCF margins as in Figure #8 below, Workday crossed the FCF positive chasm much sooner in its corporate life than Walmart. Since CY2015 (FY16), Workday has generated FCF margins better than any year in Walmart's corporate history. We note that due to changing accounting regulations and public company reporting requirements, we are unable to get an accurate FCF metric (as currently defined) for WMT until 1986, yet note that over multiple decades, free cash flow margins have been capped by CapEx requirements and limited cash generation when compared to revenue growth.

Figure 8 - FCF Margins

Source: Needham & Company LLC, Factset, Company materials

A common complaint regarding the business model of the subscription software business model is that the high relative amount of share-based compensation (SBC) given to eligible employees is not truly captured in cash flow calculations and thus is a significant liability for both shareholders and corporate profitability. If we adjust for this by using FCF less SBC expense, we note Workday's path to positive "Adjusted FCF" took six more years from 2015 to 2021 but the adjusted margin was still wider than WMT's for any year in its corporate history. We expect this Adjusted FCF metric will only increase from current levels as the company harnesses additional leverage with scale.

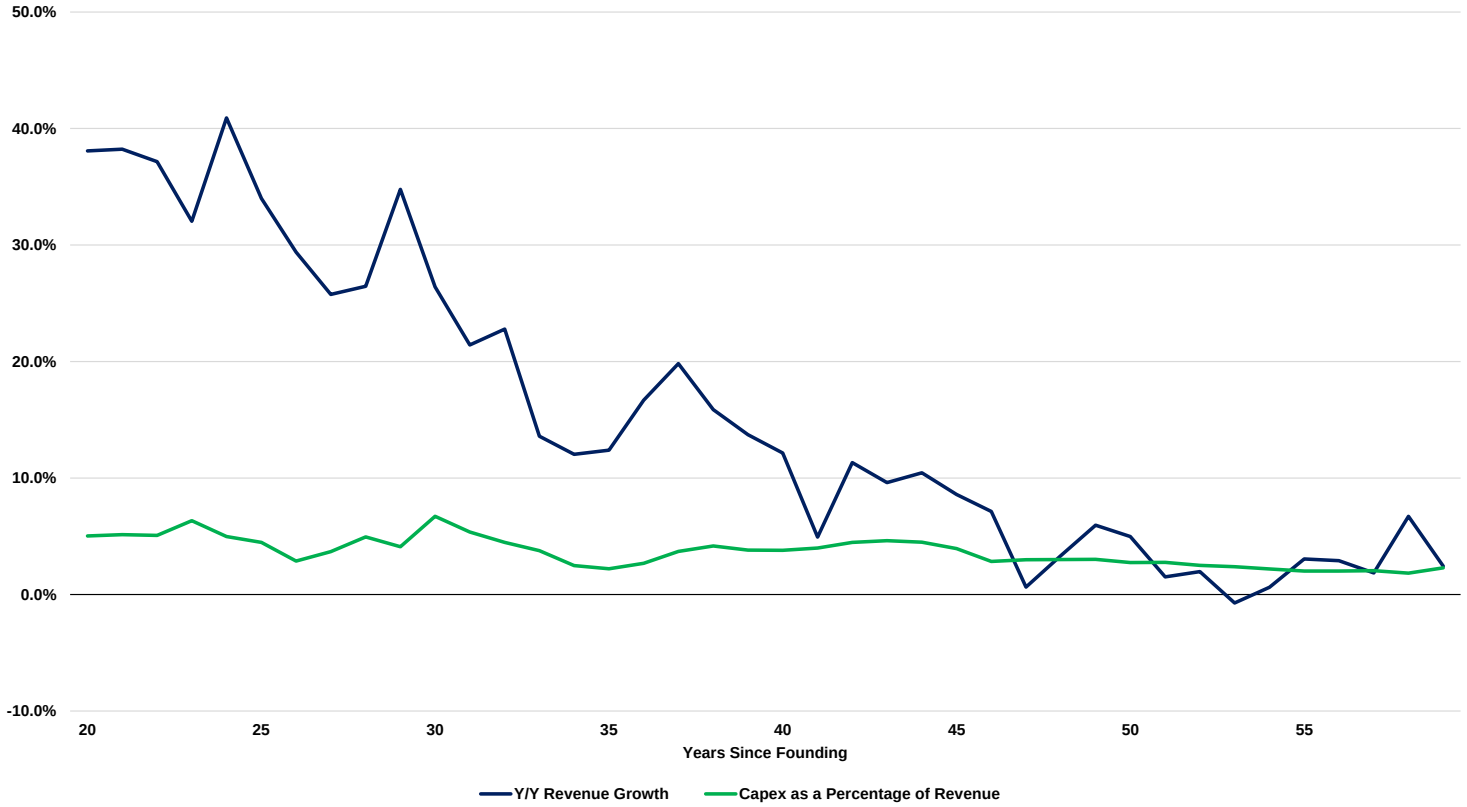
Figure 9 - FCF Margins ex-SBC



Source: Needham & Company LLC, Factset, Company materials

We believe an easy sign to identify if a business is capital intensive and relying on CapEx investment is to chart CapEx as a percent of revenues against annual revenue growth. Figure #10 below compares Walmart's annual revenue growth to its CapEx spend. Unsurprisingly, CapEx declined from 5.0% of revenue in year #20 to 2.0% in year #55, corresponding with lower store count growth.

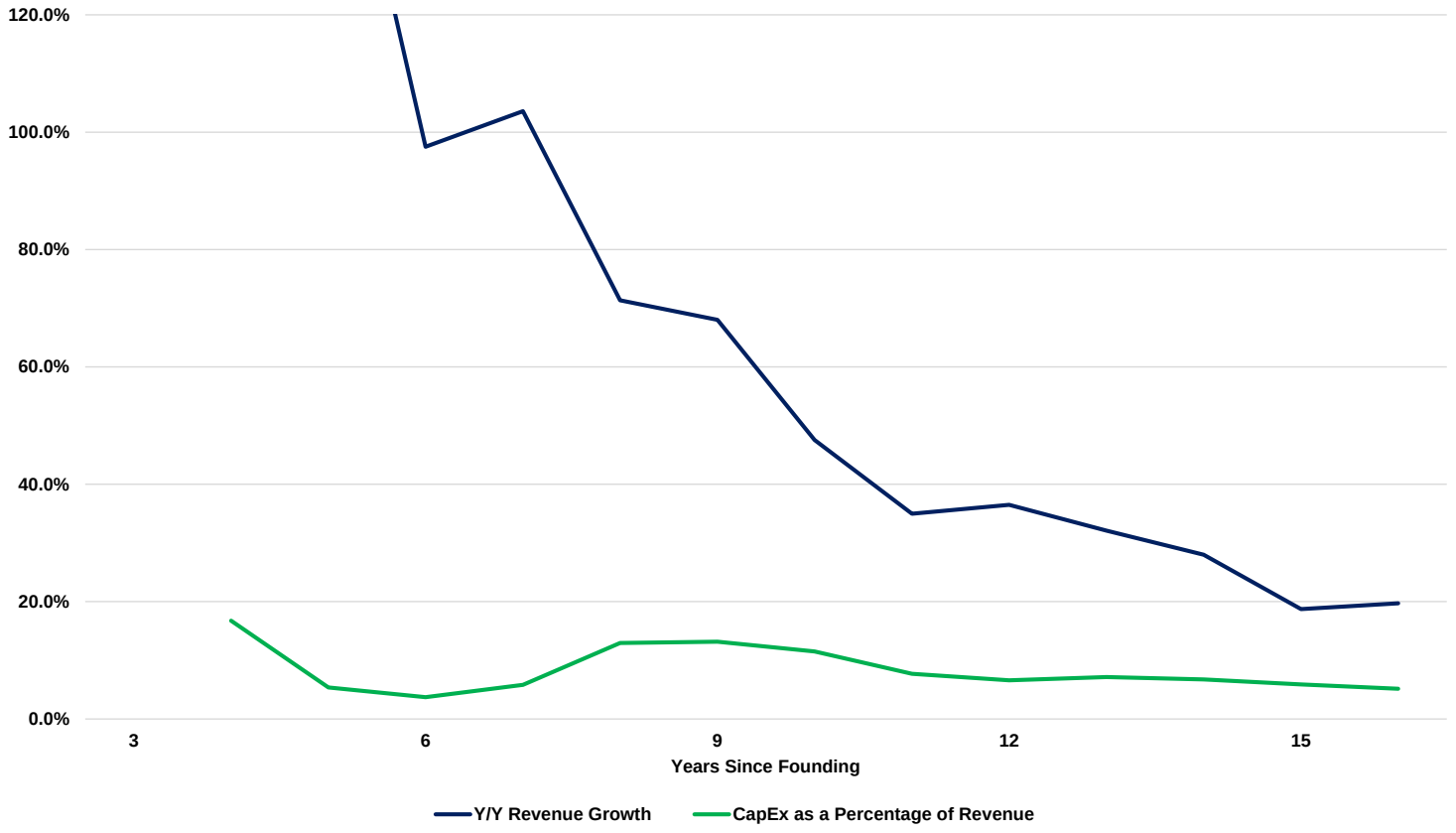
Figure 10 - WMT Revenue Growth vs CapEx as a Percent of Revenue



Source: Needham & Company LLC, Factset, Company materials

However, Workday's CapEx needs are more consistent as a percent of revenues than Walmart's because it requires constant data center hardware and investments to service new and existing customers. From effective year #5 of its corporate life to today, the company has spent roughly 6% - 7% of revenues each year on this CapEx. Note this percentage increase from FY14 - FY16 (years 8-10), we believe, from owned property investments that were not quantified the first couple of years the company made these investments before calling them out specifically in FY17, which immediately dropped CapEx back to roughly 7% of revenue.

Figure 11 - WDAY Revenue Growth vs. CapEx as a Percent of Revenue



Source: Needham & Company LLC, Factset, Company materials

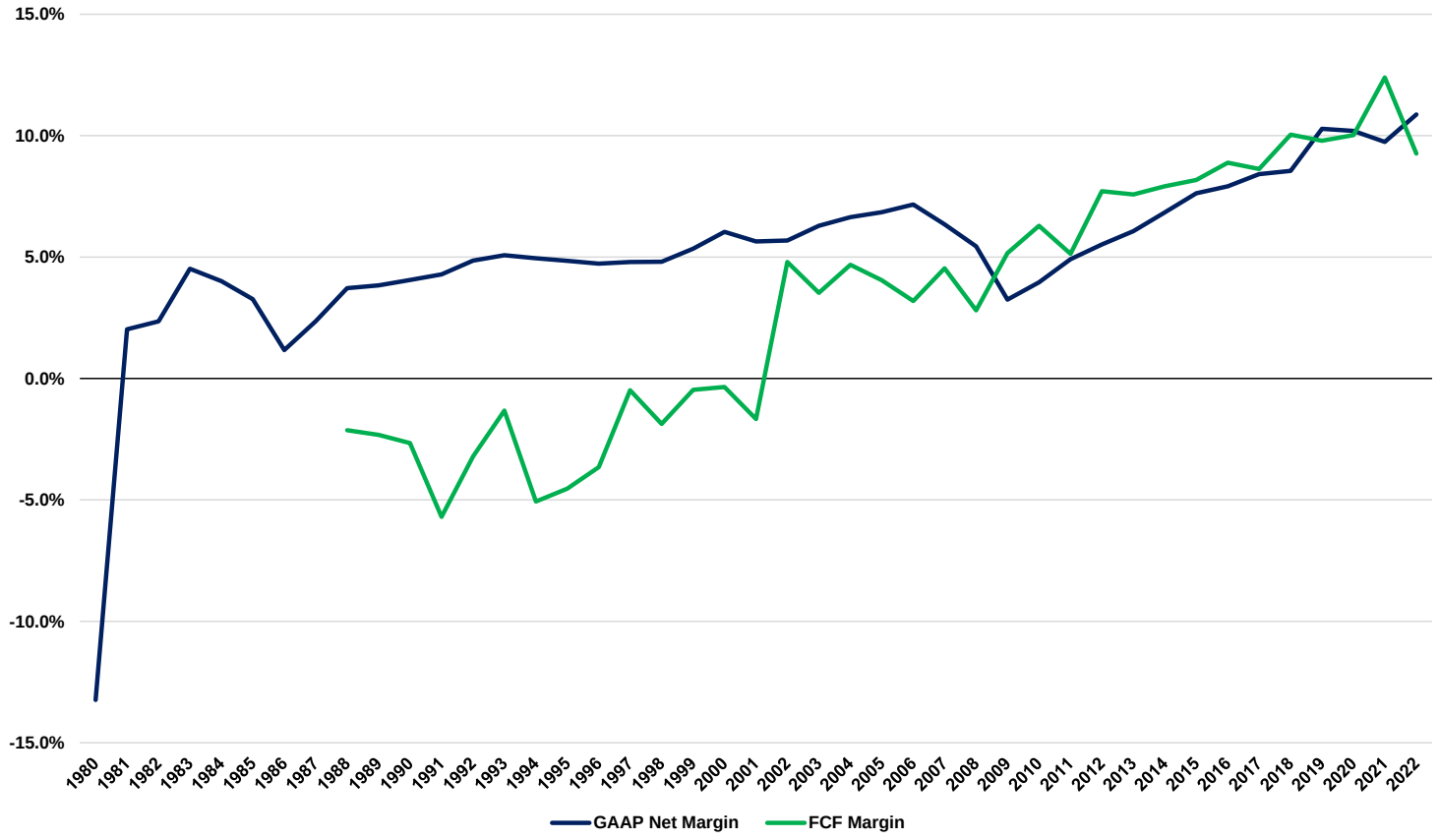
CONSUMER DISCRETIONARY - HOME DEPOT - HD

Next, we compare Workday to Home Depot (HD, not covered) in the Consumer Discretionary sector. Although the sectors' respective titles do different, we note the comparisons with Workday are nearly identical due to similarities in business model and the accounting treatment of growth investments which also is predominately CapEx based.

Home Depot generates permanent annual positive FCF 23 years post founding, positive GAAP Net Income at IPO. Home Depot went public in 1981 shortly after being founded in 1978. Home Depot did not consistently generate free cash flow until 2001 (FY02), yet has been GAAP net income positive since going public.

As the Figure below highlights, Home Depot did not generate positive FCF for a significant time frame due to investments in CapEx for store expansion through its heaviest growth years. Despite strong GAAP earnings numbers throughout the company's public company history, free cash flow margins continued to lag relative to earnings growth.

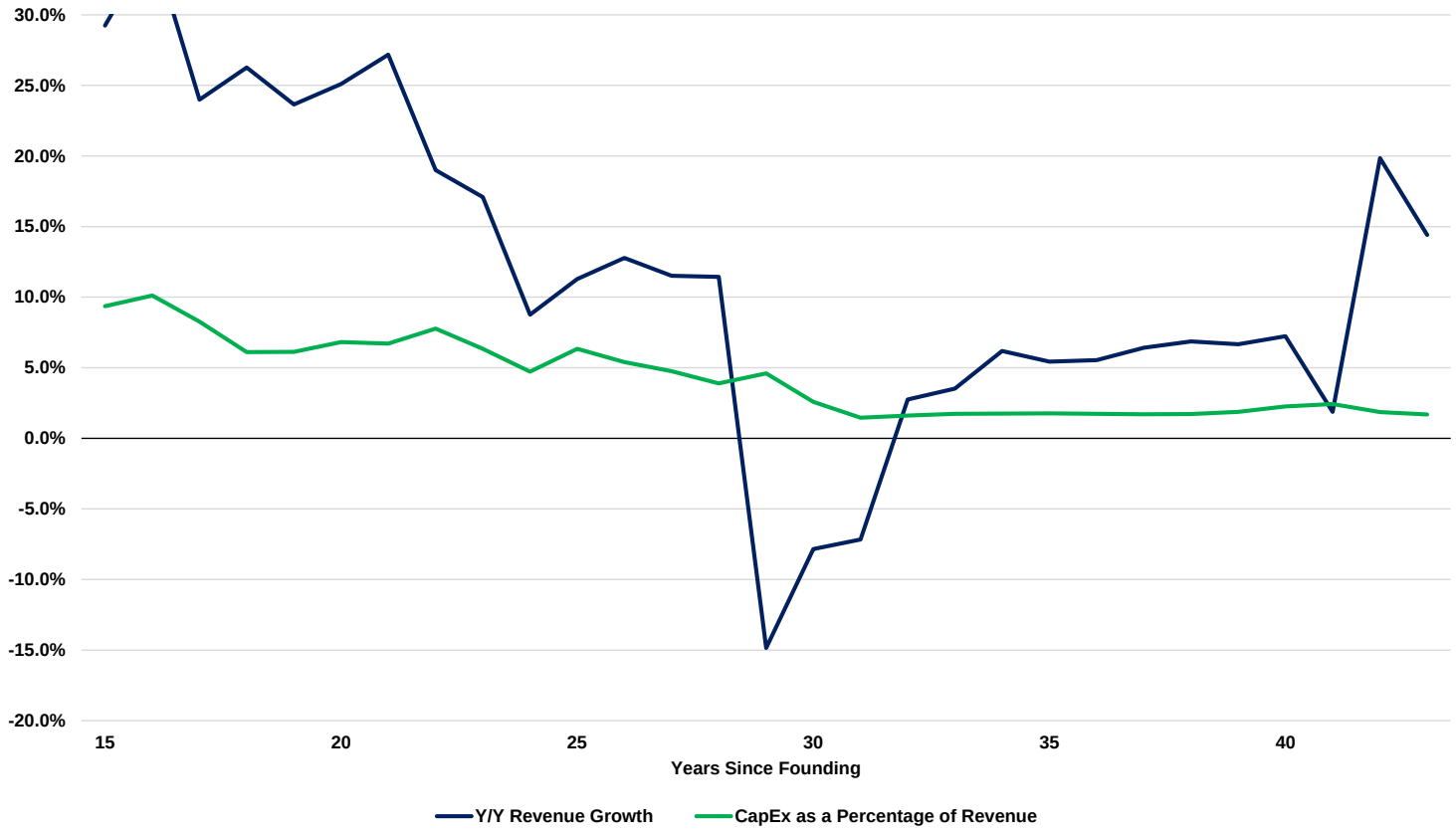
Figure 12 - HD Net and FCF Margins



Source: Needham & Company LLC, Factset, Company materials

The correlation between CapEx/PP&E spend and annual revenue growth is high at Home Depot, very similar to Walmart. As CapEx investments for new store growth slowed in the early 2000s, the company's annual revenue growth rate decelerated from the mid to upper 20% level into the low teens by 2004. CapEx spend as a percent of revenue fell from 10% in year #15 to low single digits in year #35 (and throughout the late 2010s) driving annual revenue growth to decelerate from greater than 25% to 5% for several years before the Covid-19 pandemic.

Figure 13 - HD Revenue Growth vs. CapEx as a Percent of Revenue

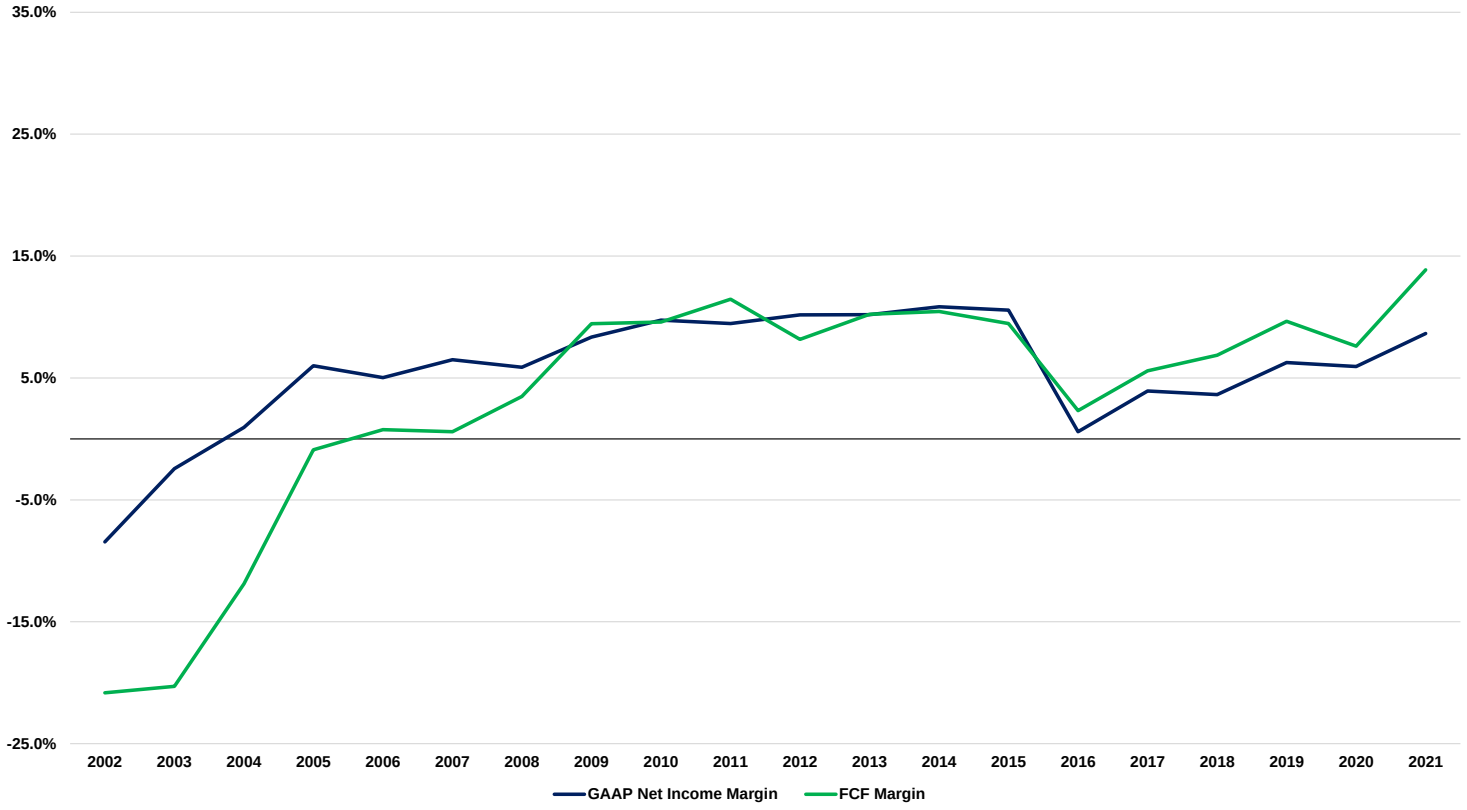


Source: Needham & Company LLC, Factset, Company materials

CONSUMER DISCRETIONARY (RESTAURANTS) - CHIPOTLE - CMG

We believe one of the best examples of a strong franchise driven business model is Chipotle Mexican Grill (CMG - NR). Chipotle was founded in 1993, yet did not go public until 2006. Chipotle demonstrated an impressive profitability turnaround in the years prior to their IPO, with a (8.4%) GAAP net margin and (20.8%) FCF margin in 2002. GAAP net income then turned positive in 2004, at a 5.0% margin in 2006. FCF margins were positive for the first time in 2006, just slightly breaking even. CapEx levels were significantly high as the company was in the early stages of growth, yet have come down to a relatively steady level just above 5% over the past ~10 years. While the CMG comparison against WDAY is not quite as strong as against WMT or HD, the company's FCF margins were significantly less than Net Margins due to store expansion.

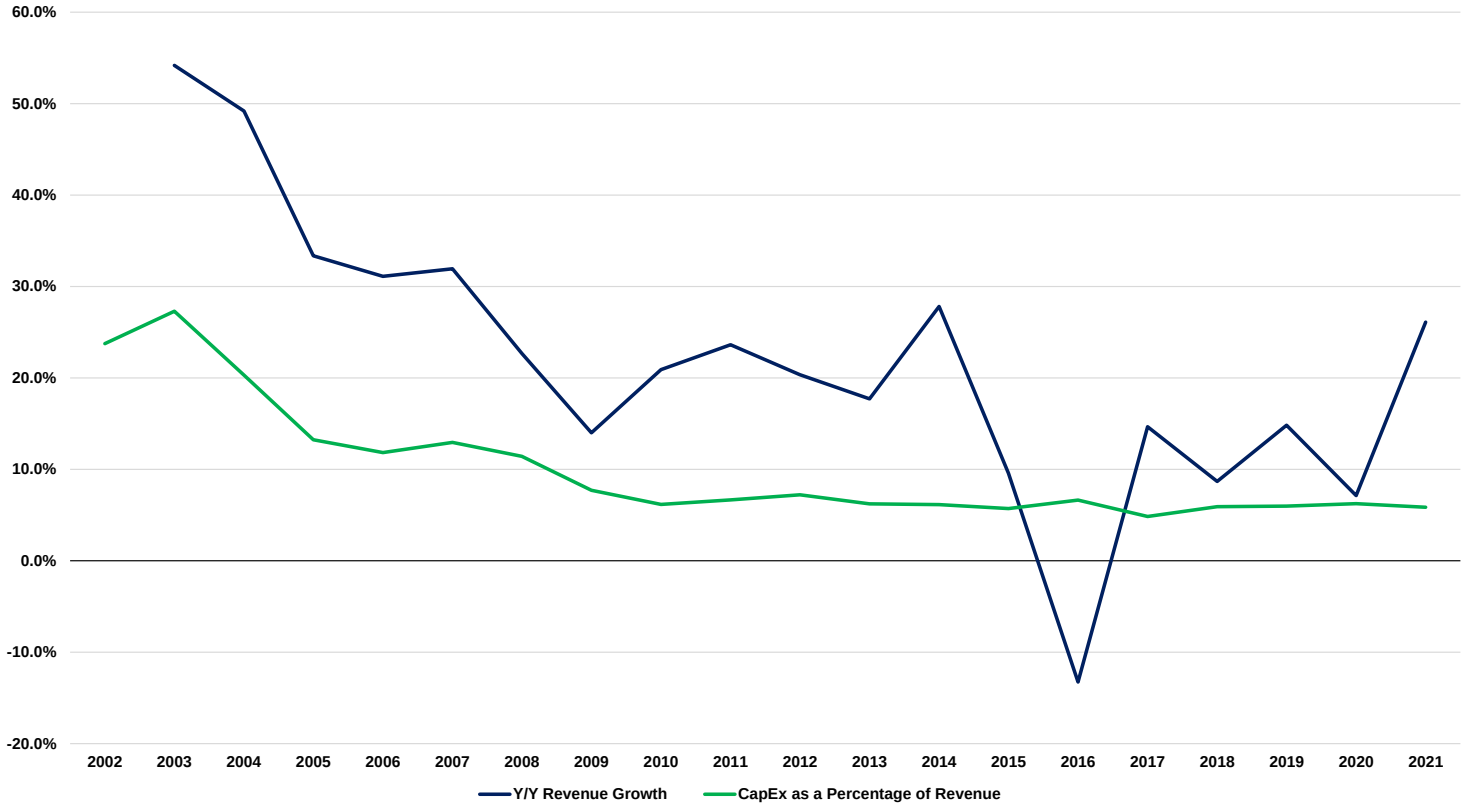
Figure 14 - CMG GAAP Net Margin vs FCF Margins



Source: Needham & Company LLC, Factset, Company materials

Figure #15 below should similar CapEx to revenue growth characteristics to other capital intensive businesses. As CapEx as a percentage of revenues have fallen (i.e. new store growth), so has revenue growth rates.

Figure 15 - CMG Revenue Growth vs CapEx as a Percentage of Revenues

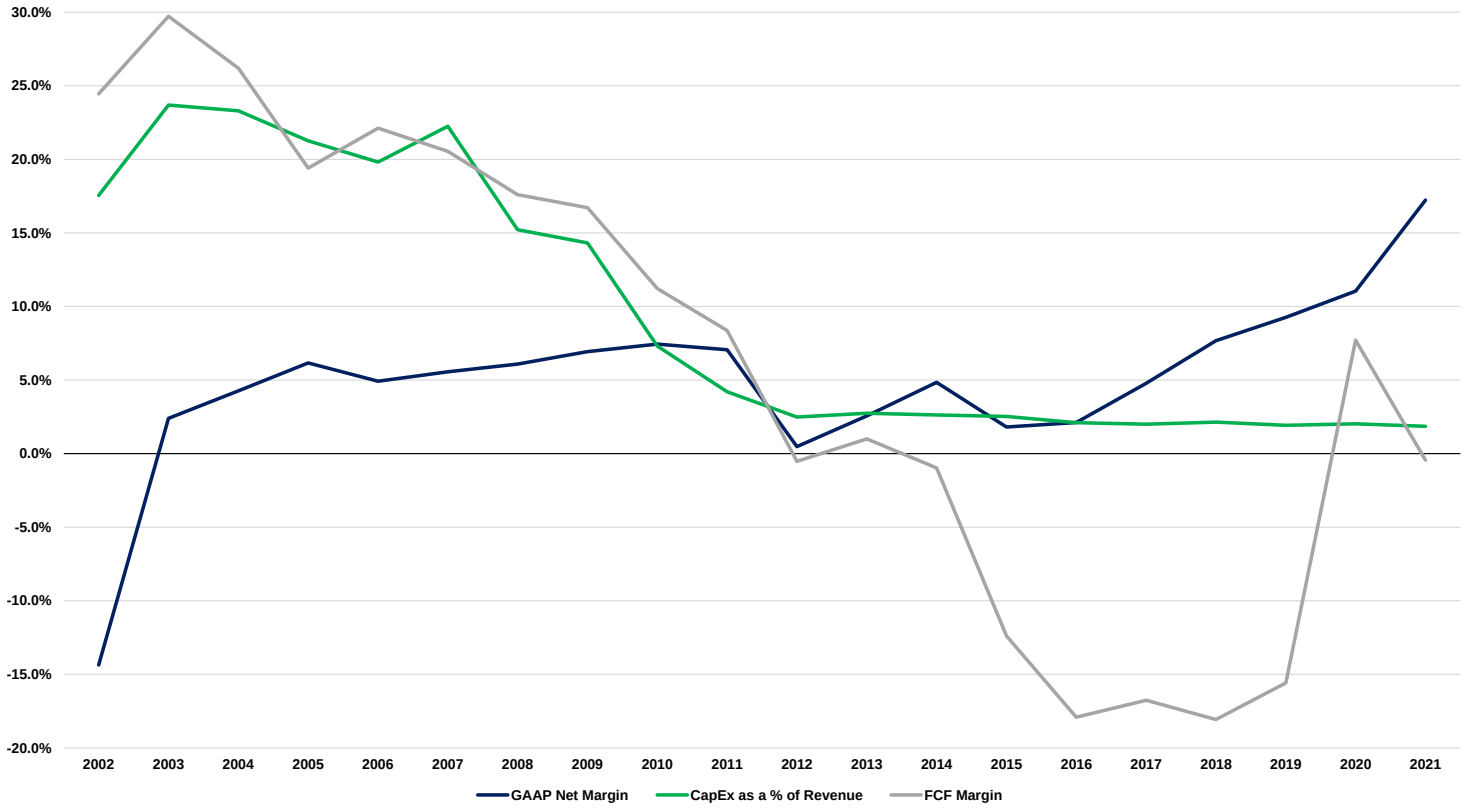


Source: Needham & Company LLC, Factset, Company materials

COMMUNICATION SERVICES - NETFLIX - NFLX

Netflix was founded in 1997 and went public in 2002, becoming one of the most popular products and best performing stocks over the following 20 years. Netflix was able to turn GAAP net income positive in 2003, just a year after IPO and six years after founding. GAAP net margins have continued to expand in impressive fashion post-2011, while CapEx as a percentage of revenue has declined to recently around 2%. However, free cash flow has been lackluster, as the business has not historically been able to produce strong operating cash flows.

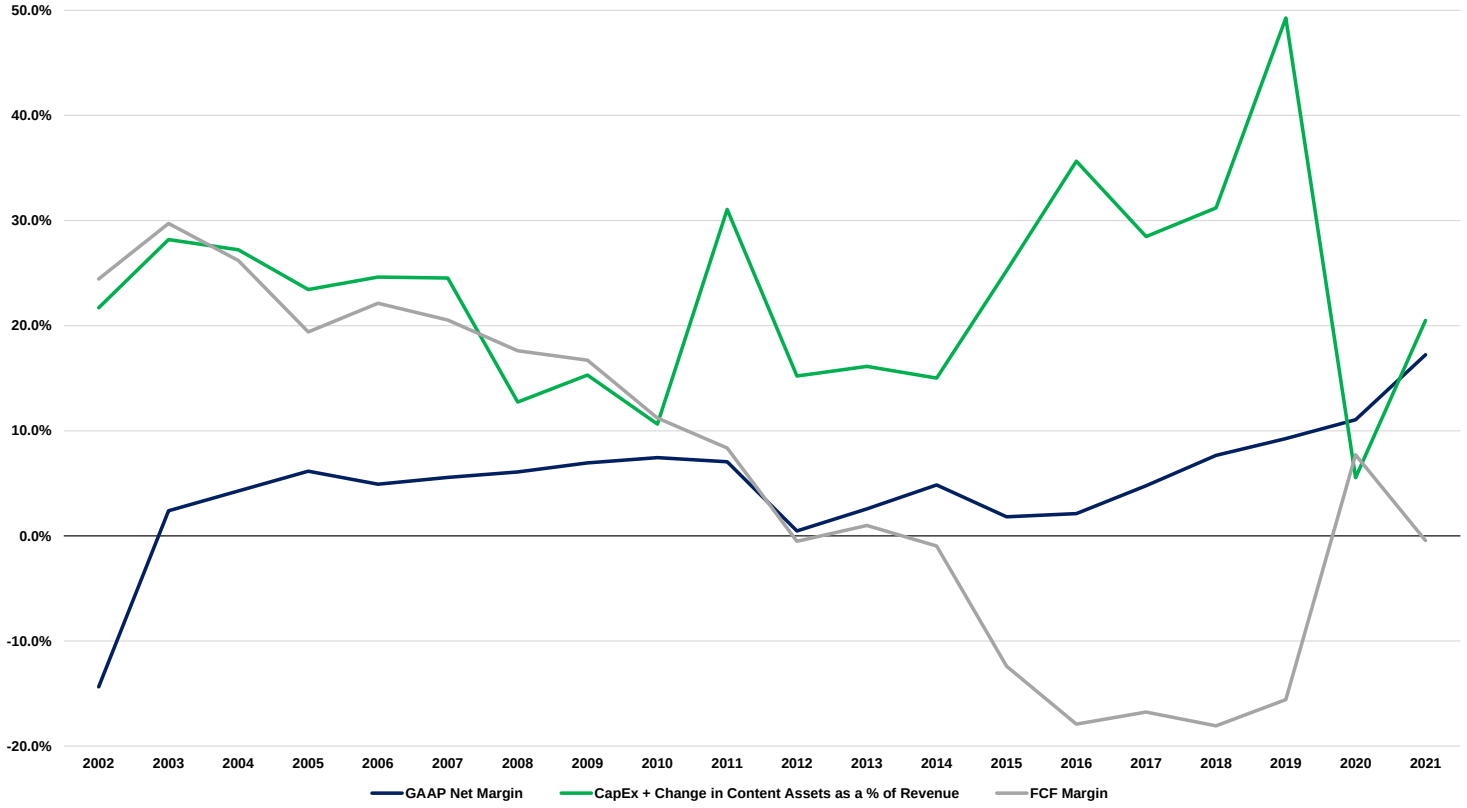
Figure 16 - NFLX Margins and CapEx



Source: Needham & Company LLC, Factset, Company materials

However, this quick net income vs FCF comparison does not accurately describe the growth investment picture at Netflix. Beginning in the early 2010s, the company started developing and creating its own content and capitalizing the expenses as a separate line item. If we add these expenses to CapEx to properly reflect the company's growth investments as in Figure #17 below, it is easily to understand the primary driver between Net Margins and FCF margins throughout the 2010s is content capitalization, and while profitable today, capitalization is a primary driver of why the company is profitable today.

Figure 17 - NFLX Capitalized Content Impact on FCF Margins



Source: Needham & Company LLC, Factset, Company materials

The Netflix operating model also looks a lot like the prior capital intensive business models. Declining CapEx investments were quickly followed by declining revenue growth but increased capital investments also led to a quick increase in revenue growth rates.

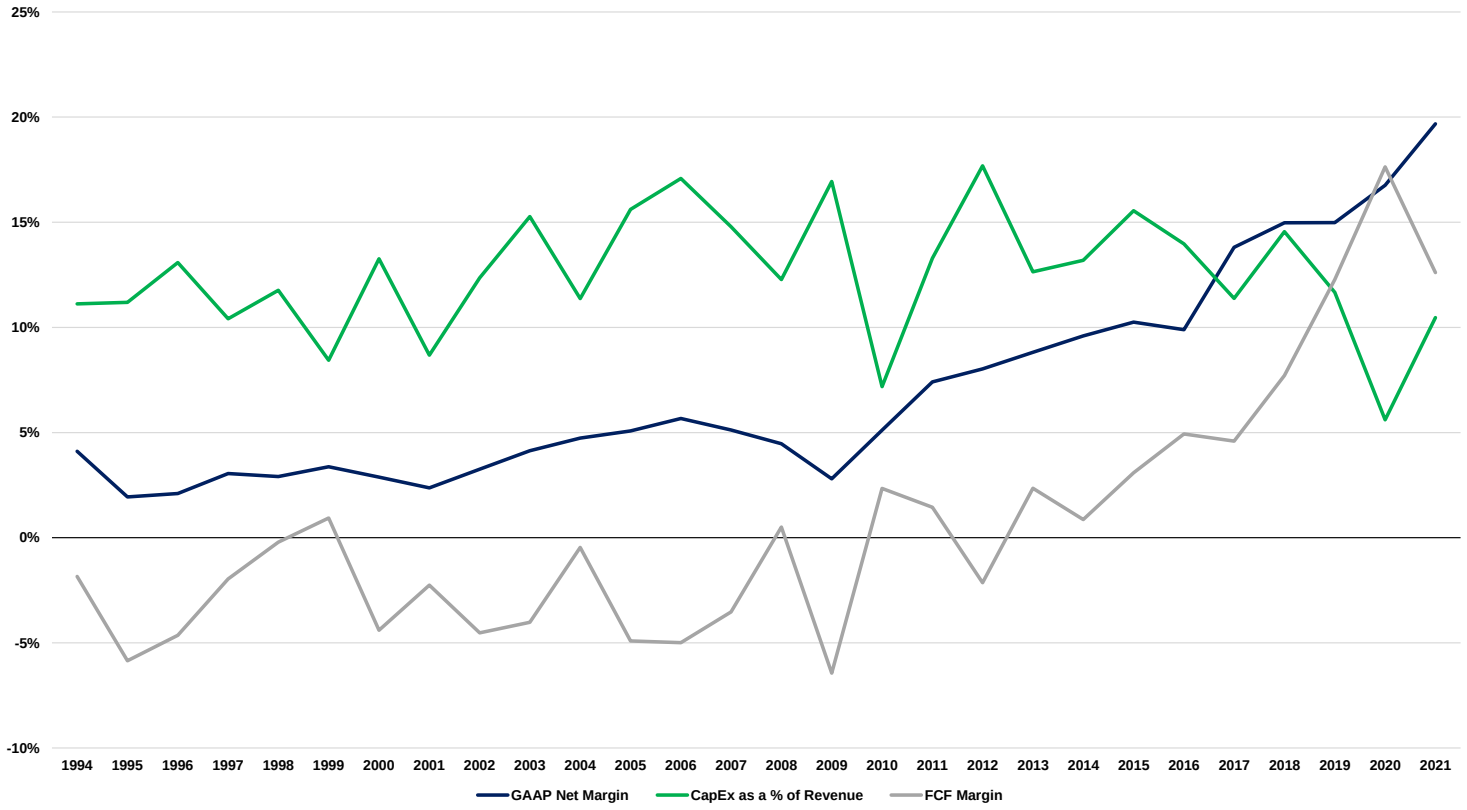
Figure 18 - NFLX Revenue Growth vs. CapEx as a Percent of Revenue

Source: Needham & Company LLC, Factset, Company materials

INDUSTRIALS - OLD DOMINION FREIGHT LINE - ODFL

A mature, capital intensive business we investigate is Old Dominion Freight Line (ODFL, not rated). ODFL was founded in 1934 and went public in 1991. GAAP net income has been positive every year since the company's IPO, yet the company did not generate consistent free cash flow until 2013, despite a few slightly positive years along the way. This cap on free cash flow generation for all of those years comes due to high capital expenditures, which have been generally over 10% over time. FCF margins have been trending up since flipping positive in 2013, and jumped from 7.7% to 12.3% in 2019 before reaching an all-time high of 17.6% in 2020 as CapEx dropped to just 5.6% of revenue during the same year. In 2021, CapEx was back near historical levels of 10.5% of revenue, while FCF margins dipped back to 12.6%, albeit well above the historical FCF level.

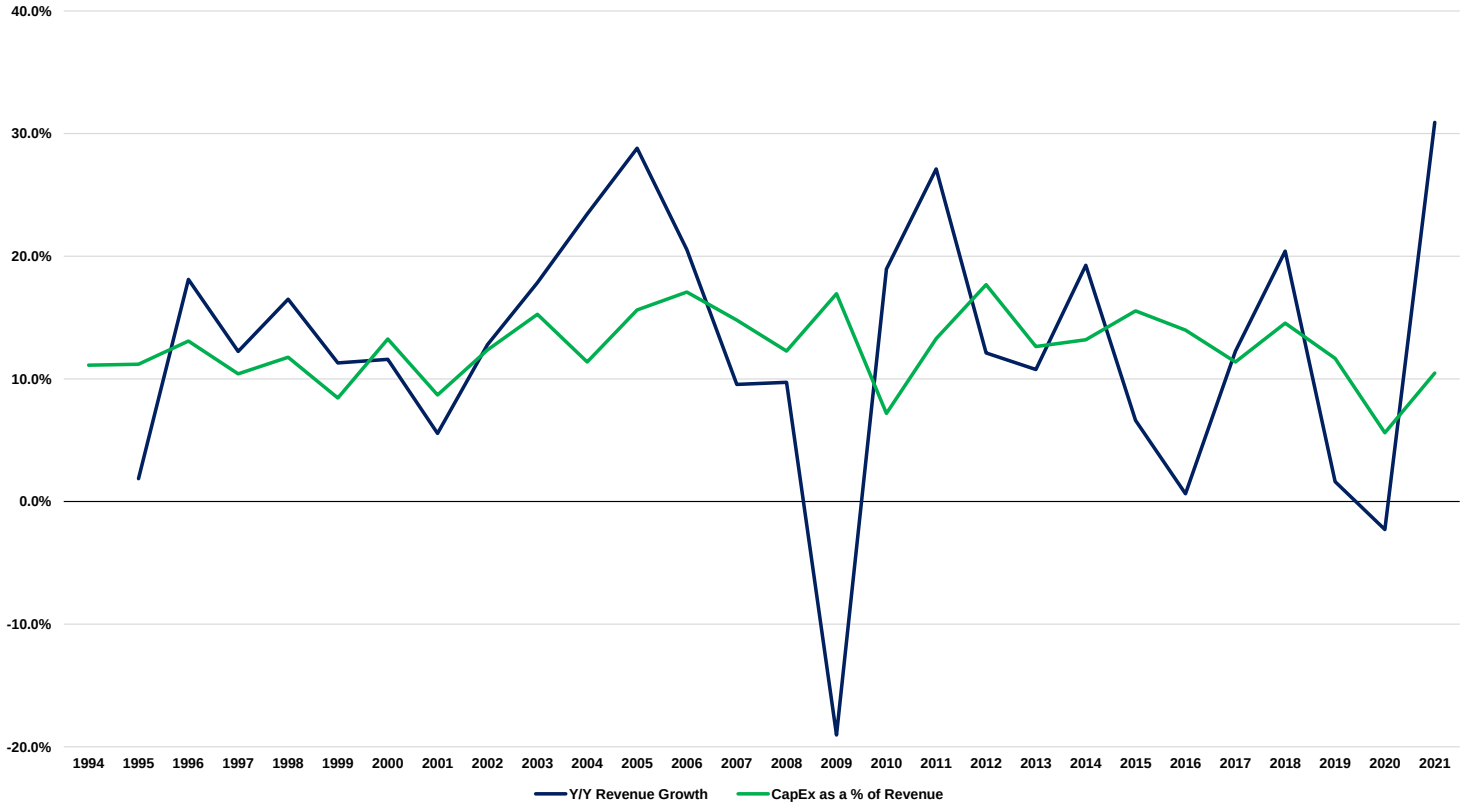
Figure 19 - ODFL Margins and CapEx



Source: Needham & Company LLC, Factset, Company materials

Analyzing CapEx investments versus revenue growth rates at ODFL also point to a similar capital intensive model. Declining CapEx investments were quickly followed by declining revenue growth but increased capital investments also led to a quick increase in revenue growth rates.

Figure 20 - ODFL Revenue Growth vs. CapEx as a Percent of Revenue



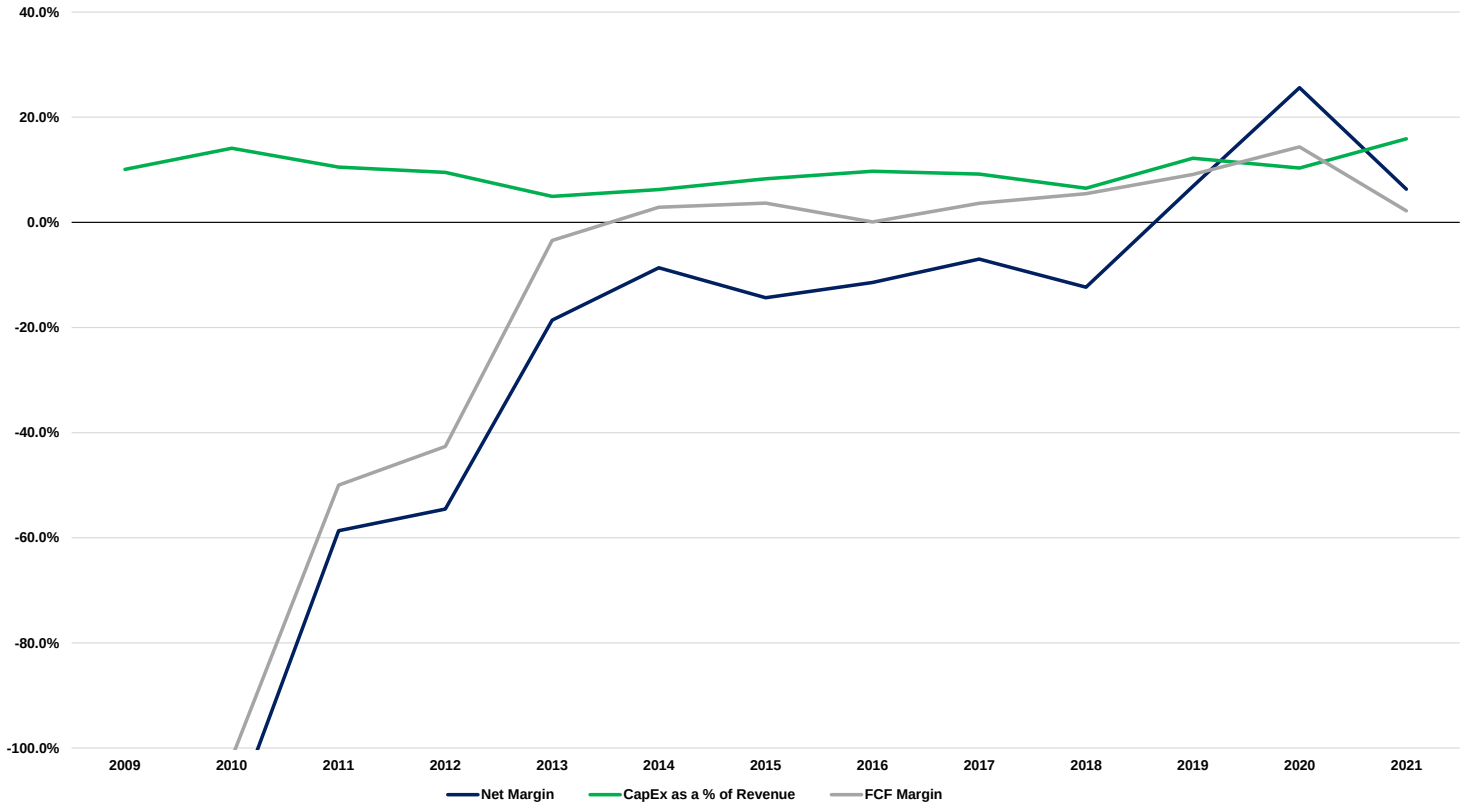
Source: Needham & Company LLC, Factset, Company materials

HEALTHCARE - DEXCOM - DXCM

Looking into a somewhat unique healthcare equipment business model, we examine Dexcom (DXCM, not rated), which was not generating revenue at the time of its IPO. Founded in 1999, Dexcom didn't go public until 2005 and didn't generate revenue until 2006. After the company began producing revenue, it was able to achieve nearly a 60% CAGR from 2006 to 2021, supported by strong tailwinds in the diabetes space and a revolutionary product portfolio. Dexcom wasn't GAAP Net Income positive until 2019, yet it started to generate positive free cash flow in 2013.

The Dexcom business model is actually quite similar to the Workday model. Most growth investments around R&D are expenses as incurred which is apparent in the close correlation between Net Income and FCF margins and the generally consistent cadence of FCF as a percent of revenue spend. The company was able to generate FCF before generating positive GAAP Net Income as it gained leverage over prior product R&D expenses.

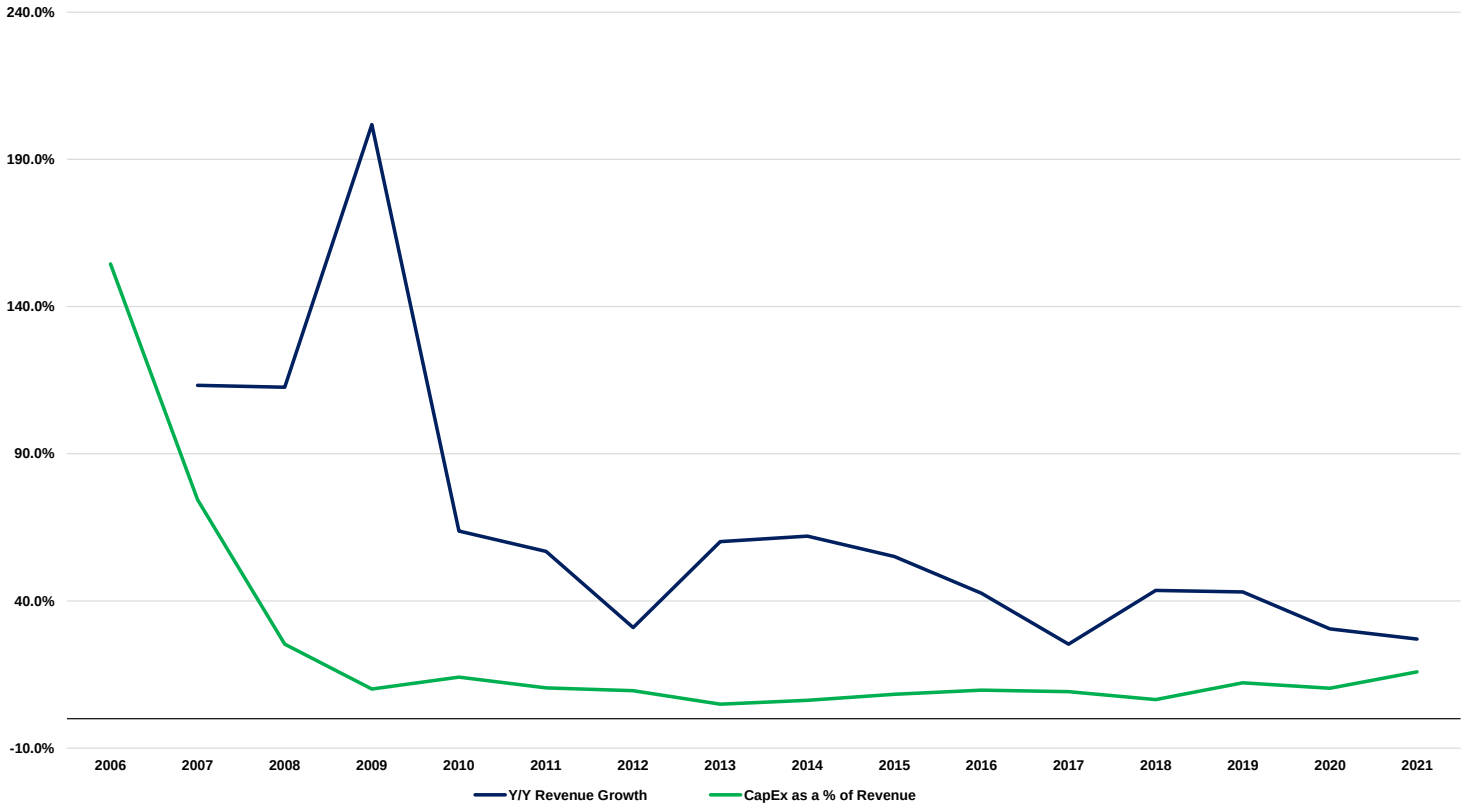
Figure 21 - DXCM Margins and CapEx



Source: Needham & Company LLC, Factset, Company materials

Analyzing DXCM's CapEx spend versus its revenue growth rate does not have the same correlation to other capital intensive business and is more akin to the Workday investment model. While CapEx as a percent of revenues has been fairly consistent over its life, changes in CapEx spend have not driven immediate changes to revenue growth, either higher or lower.

Figure 22 - DXCM Revenue Growth vs. CapEx as a Percent of Revenue



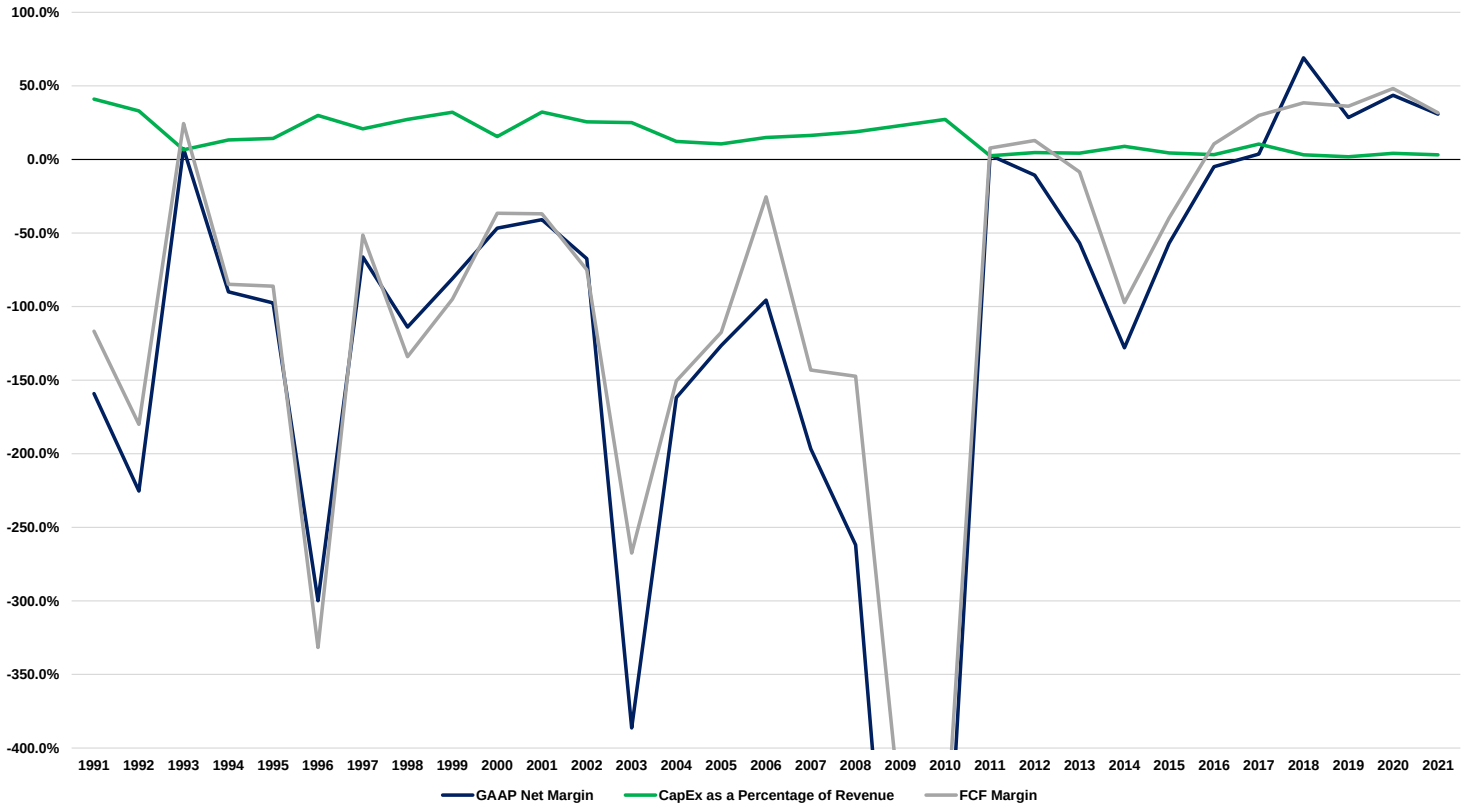
Source: Needham & Company LLC, Factset, Company materials

HEALTHCARE (BIOTECH) - VERTEX PHARMACEUTICALS - VRTX

The Biotech operating model is also quite similar to the subscription software company operating model. We analyzed Vertex Pharmaceuticals (VRTX - Rated Hold by Joseph Stringer) was founded in 1989 and went public in 1991. The biotech company invested heavily during their first 22 years, generating low amounts of revenue with CapEx spend averaging more than 20% of revenue over that time period. This resulted in significantly negative margins every year but two until 2016, where the company was able to flip both GAAP net income and FCF margins positive and generate strong profitability growth over the following years. This comes following a notable decline in relative CapEx spend, which fell below 5% in 2011 as the company began to generate significant revenue for the first time, growing total revenue 884% Y/Y.

While the company is gaining leverage over CapEx investments similar to more capital intensive sectors, the company's core growth investments are R&D operating expenses, which are expensed as incurred and not capitalized. A pharmaceutical company's ability to generate revenue from new R&D is the direct, corresponding driver to GAAP Net Income and positive FCF, similar to the viability of a software application.

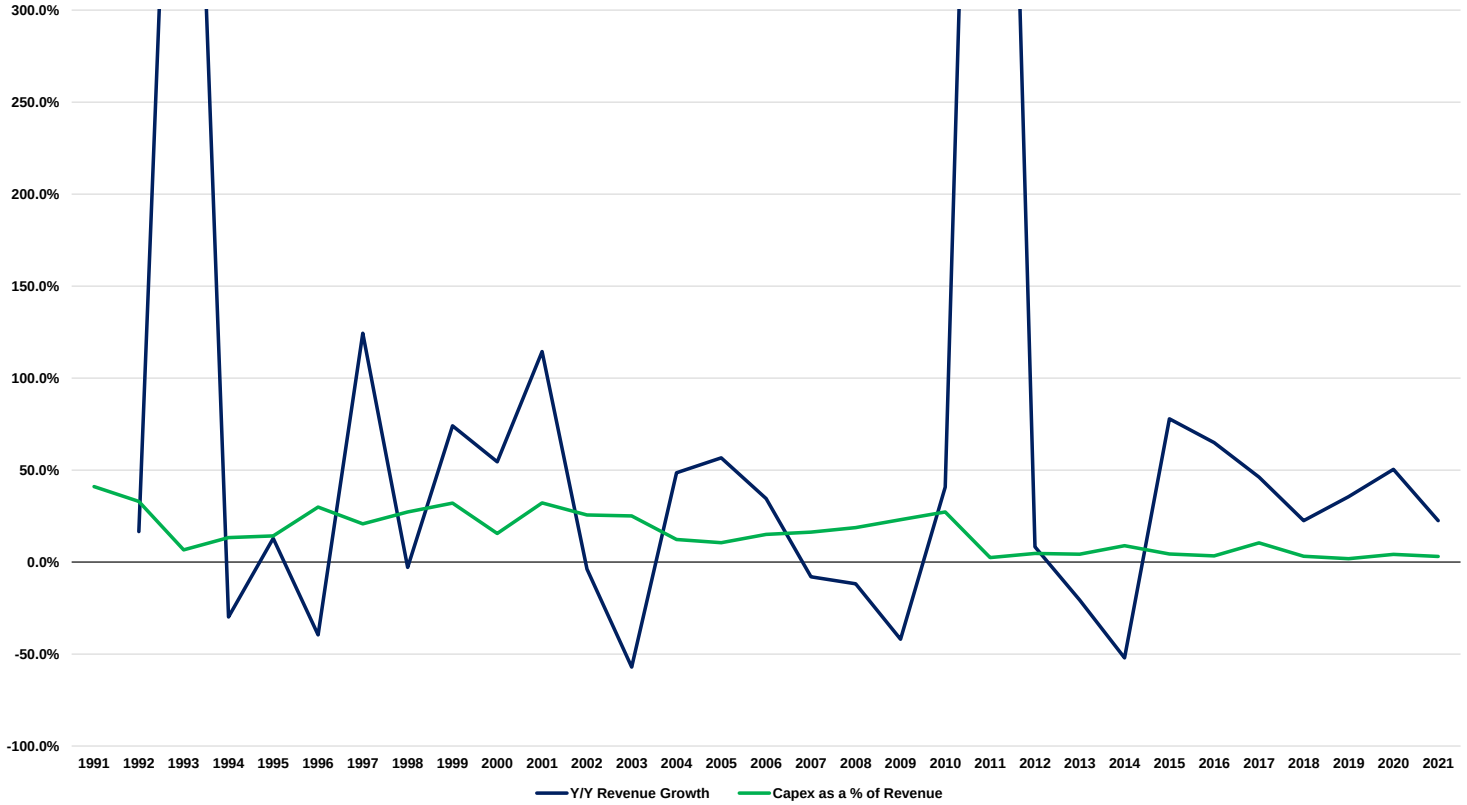
Figure 23 - VRTX Margins and CapEx



Source: Needham & Company LLC, Factset, Company materials

Analyzing VTRX's CapEx spend versus its revenue growth rate does not have the same correlation to other capital intensive business and is more akin to the Workday investment model. While CapEx as a percent of revenues has been fairly consistent over its life, changes in CapEx spend have not driven immediate changes to revenue growth, either higher or lower.

Figure 24 - VTRX Revenue Growth vs. CapEx as a Percent of Revenue

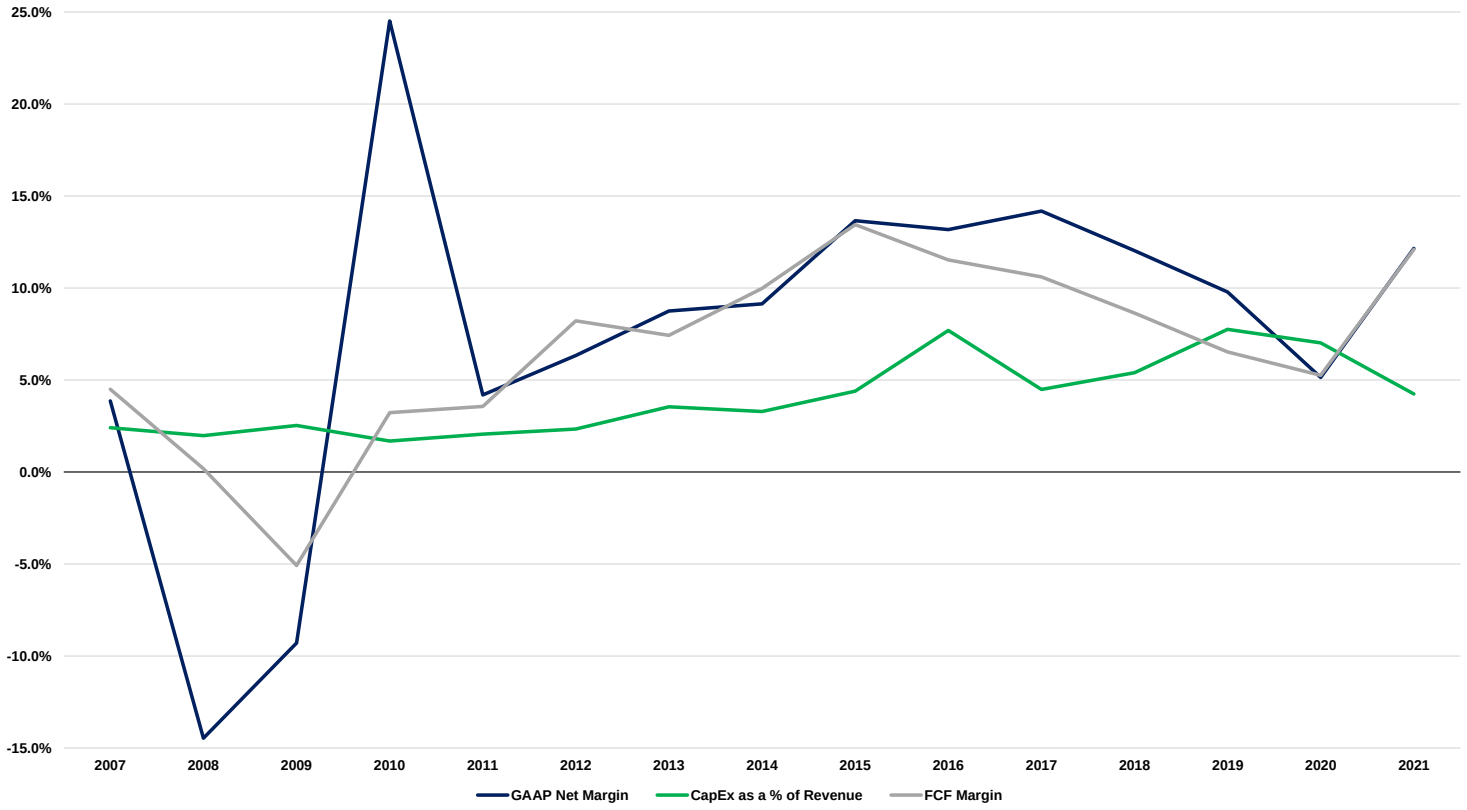


Source: Needham & Company LLC, Factset, Company materials

MATERIALS - LYONDELLBASELL - LYB, NOT RATED

LyondellBasel was founded in 2007 and went public in 2010. The company has generated positive GAAP net income in every year but two as a public company. CapEx spend has been generally limited, while free cash flow margins have been positive every year but 2009. We note that despite the strength of this model, its FCF margin regularly underperforms Workday's. We see this model as interesting due to the fact that despite low CapEx, LYB has been generally capped in regard to FCF production as they have matured as a public company.

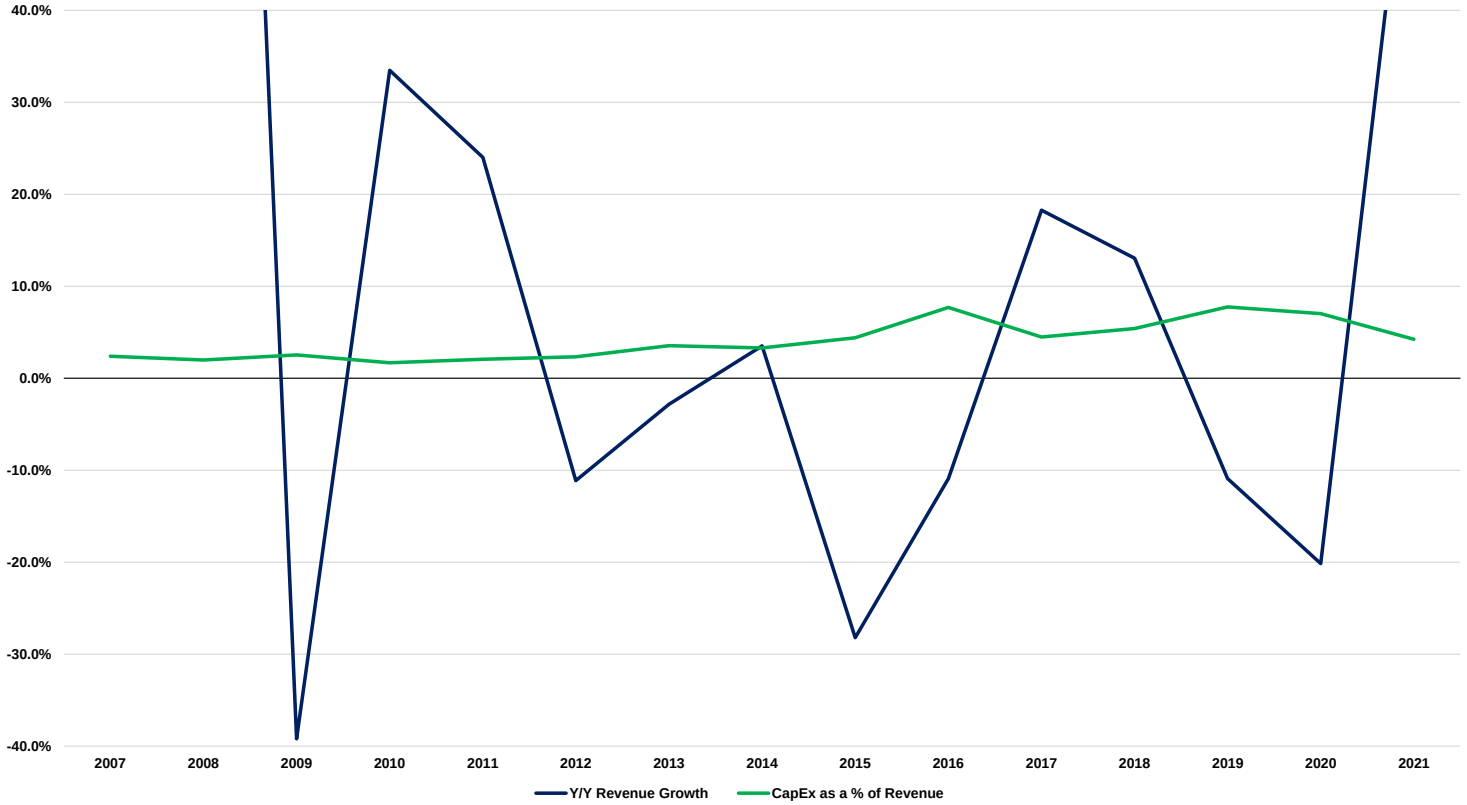
Figure 25 - LYB Margins and CapEx



Source: Needham & Company LLC, Factset, Company materials

Analyzing LYB's CapEx spend as a percent of revenues versus revenue growth is a bit inconclusive with LYB. Figure #26 below does not highlight a tight correlation between CapEx spend and revenue growth but does seemingly have a marginal lagging impact. Of all the operating models analyzed in this report, we find LYB's the least conclusive in terms of understanding the exact impact growth investments have on a company's operating model.

Figure 26 - LYB Revenue Growth vs. CapEx as a Percent of Revenue

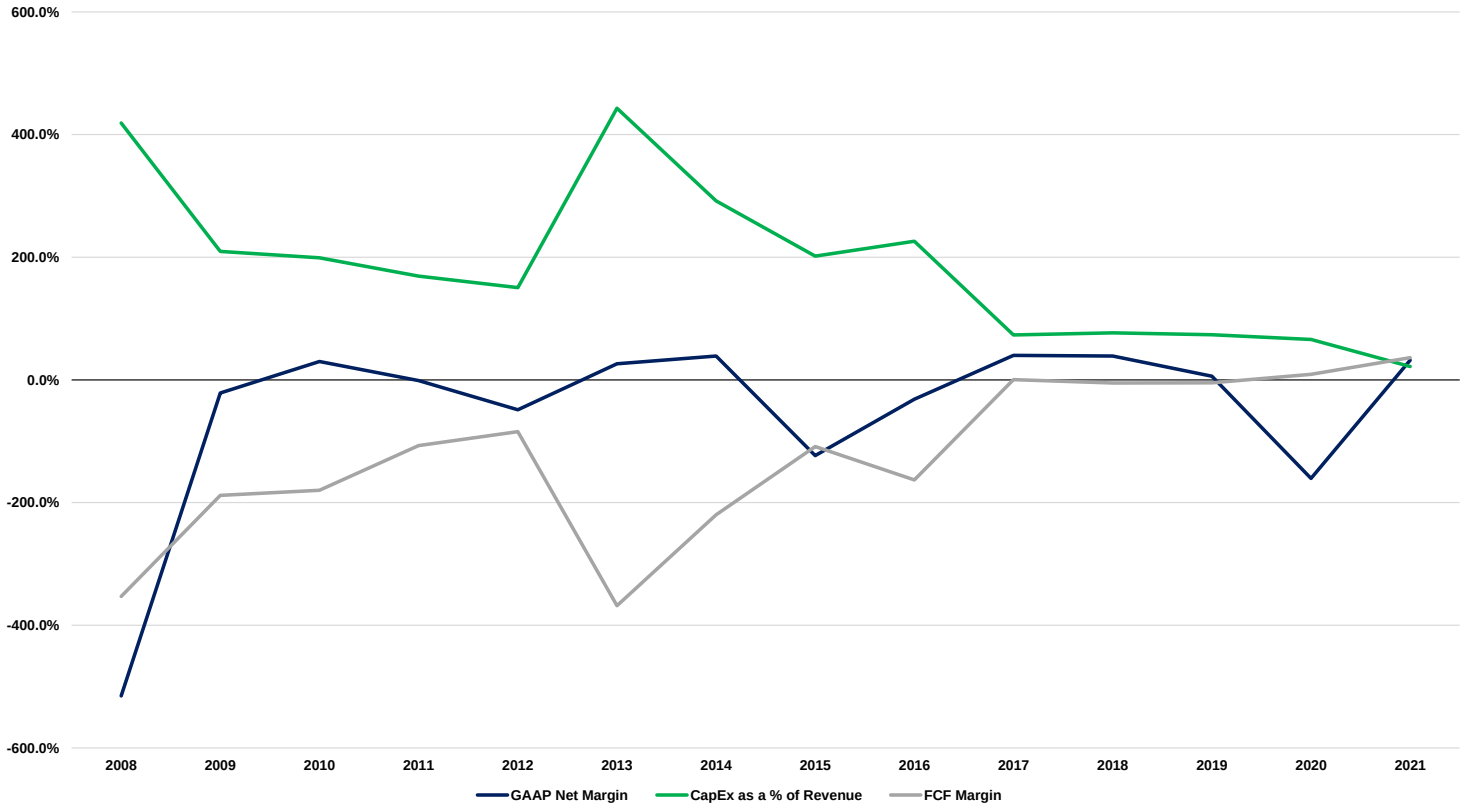


Source: Needham & Company LLC, Factset, Company materials

ENERGY - DIAMONDBACK ENERGY - FANG, NOT RATED

Diamondback Energy was founded in 2007 and went public in October 2012. Despite a strong cyclical aspect, the company has managed to produce positive GAAP net income four out of the last five years and six out of their ten reported years as a public company. However, significantly high CapEx requirements limit the company's ability to produce free cash flow, which it did not show until 2020, or 12 years after its founding. As shown in Figure 27 below, CapEx has been the main hindrance of FCF production, as operating cash flows have been strong since the company was founded.

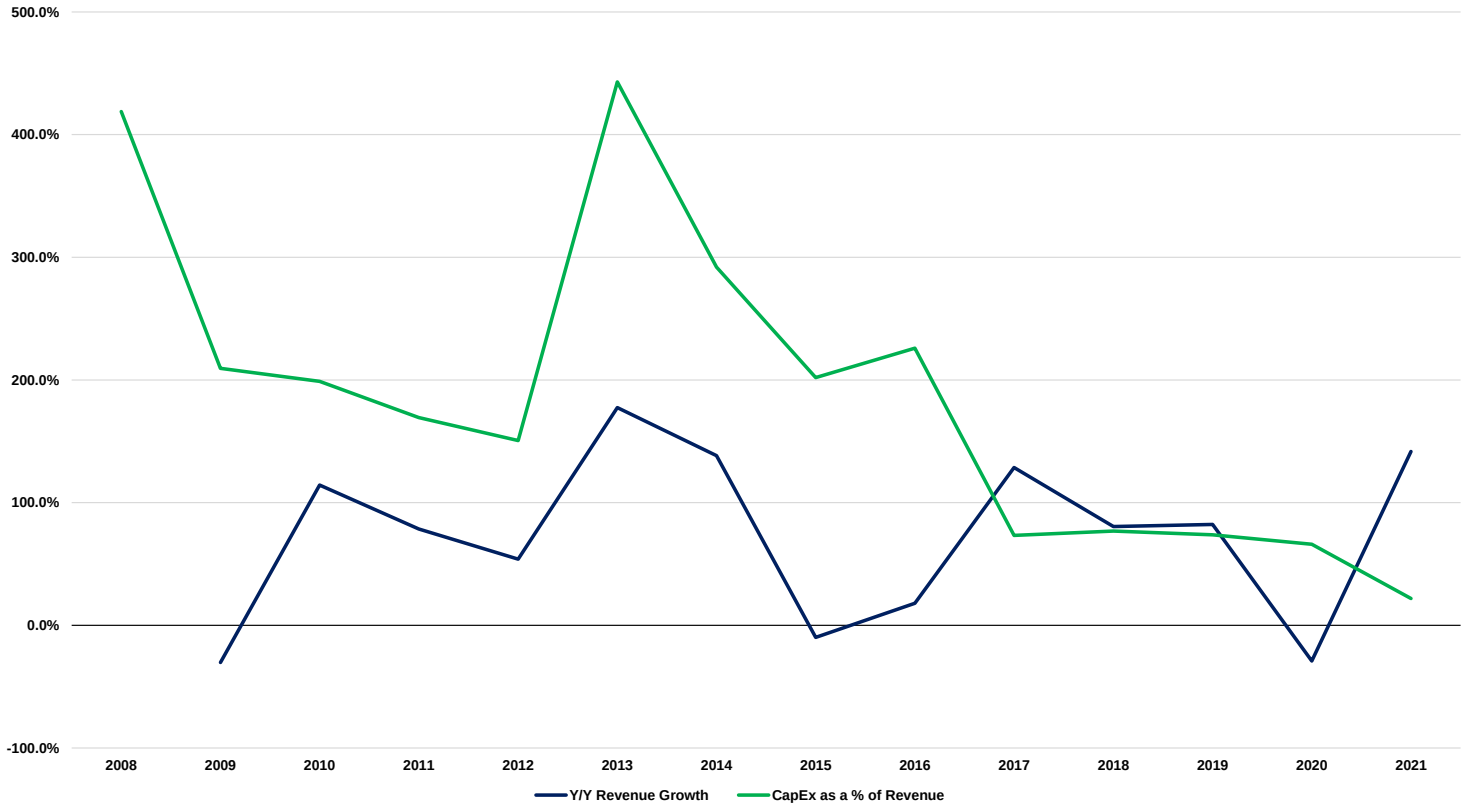
Figure 27 - FANG Margins and CapEx



Source: Needham & Company LLC, Factset, Company materials

Analyzing CapEx investments versus revenue growth rates at FANG also point to a similar capital intensive model. Declining CapEx investments were quickly followed by declining revenue growth but increased capital investments also led to a quick increase in revenue growth rates.

Figure 28 - FANG Revenue Growth and CapEx as a Percent of Revenue



Source: Needham & Company LLC, Factset, Company materials

UTILITIES - NRG ENERGY - NRG, NOT RATED

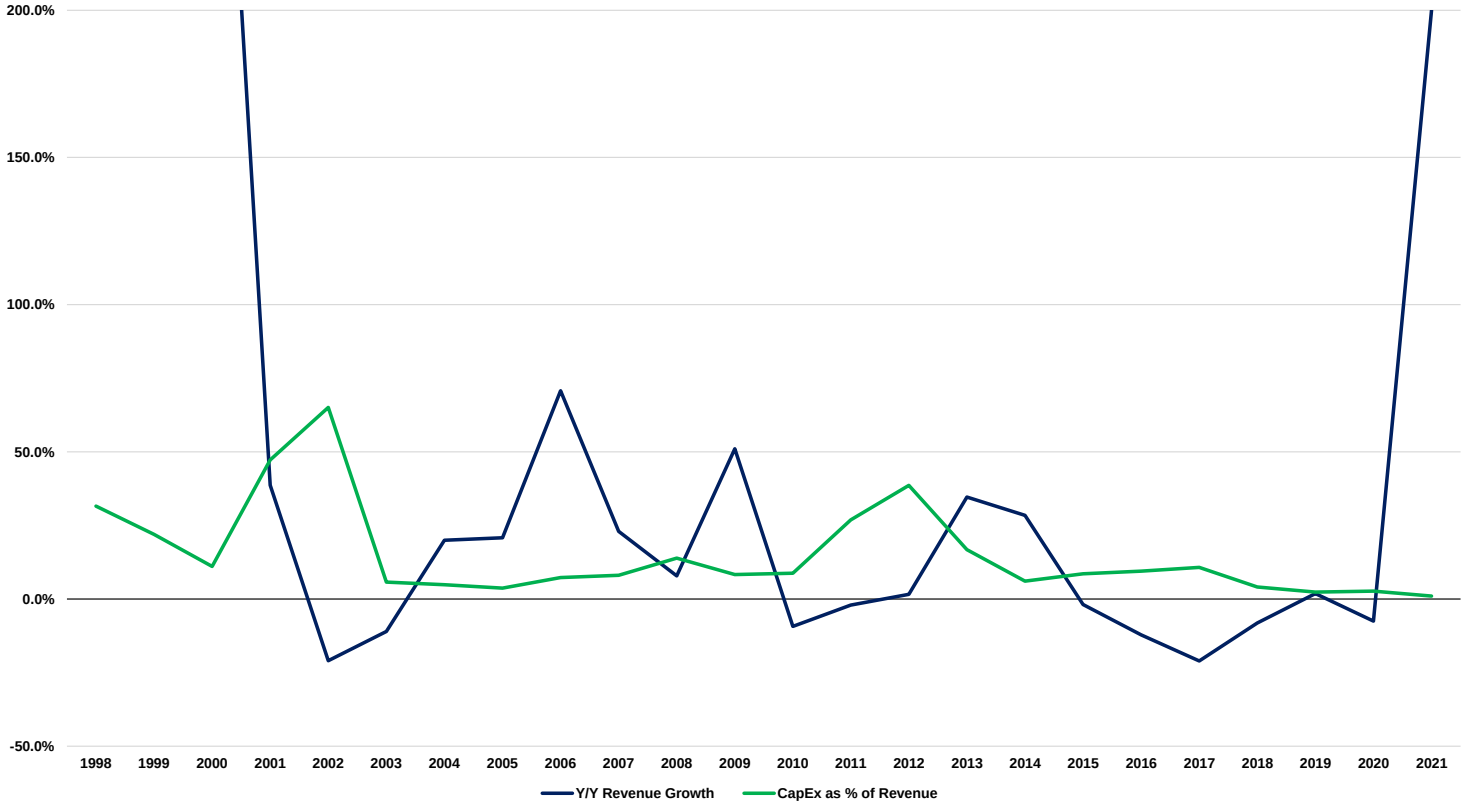
We note that the utilities sector as a whole has generally not been able to produce consistent free cash flow. Our example of NRG energy was founded in 1989 as a subsidiarity of Northern States Power Company, ultimately going public in 2000 in a spin-off. We note that NRG has done a solid job in lowering capital expenditures as a percentage of revenue as they have grown and become a more mature independent company, however free cash flow margins have not reflected this decreased relative spend, and occasional spikes in CapEx have taken a toll on FCF margins in multiple years, including 2002 and 2012.

Figure 29 - NRG Margins and CapEx

Source: Needham & Company LLC, Factset, Company materials

Analyzing CapEx spend as a percent of revenues versus revenue growth is a bit inconclusive with NRG. Figure #30 below does not highlight a tight correlation between CapEx spend and revenue growth but does seemingly have a marginal lagging impact. Of all the operating models analyzed in this report, LYB's would be the least conclusive in terms of understanding the exact impact growth investments have on a company's operating model.

Figure 30 - NRG Revenue Growth vs. CapEx as a Percent of Revenue



Source: Needham & Company LLC, Factset, Company materials

RISKS

BILL

- **Interest rate and general market risk.** The company is exposed to fluctuations in interest rates and if rates decline, interest from customer funds will also likely decline despite the number of customers growing. While the company invests in highly liquid, investment grade securities, customer fund assets are subject to general market, interest rate, credit, foreign exchange and liquidity risks.

- **Competitive threats.** Primary competition is from mid-market focused firms that move down market to capture customers. While we do not believe any of these companies has a simplified self-service model to attract new customers that will threaten Bill.com, greater investment by competitors in the future could lead to increased competition.

- **Transition to Public Cloud.** Bill.com is currently transitioning from Private to Public Cloud with Amazon Web Services. While we anticipate a smooth transition given the history of customers transitioning to the Public Cloud, any disruption in service could impact agreements with customers.

- **Risks related to partnerships.** A significant portion of new customers is derived through partners including Accounting Firms and Financial Institutions. While FIs have agreements related to minimum payments including incentives to drive customer growth, Accounting Firms do no operate with these agreements, and we do not believe any minimum agreement would be sufficient to meet investor expectations. In addition, FIs have greater financial resources and could enter the market with proprietary solutions at any time. However, we believe developing these solutions would require significant outlays of capital and finding qualified talent to manage. Partners may choose not to renew contracts as in late 2018, a FI ended their relationship resulting in a loss of 5,000 customers.

- **Relationship with Intuit.** The company's relationship with Intuit is an important driver of new and existing business including thousands of existing customers. While they do not drive a material or 10% portion of revenue, we believe their customer base remains

an important driver of new customer growth. If the terms of the relationship were to change or end, we believe Bill.com's growth expectations would be impacted. The terms of their agreement with Intuit do no prohibit their own development of platform or solution to manage back-office automation. While we do not currently expect the company to enter the market with a solution, this could change over time.

- **Integrations with popular accounting software packages.** For Bill.com to offer a complete solution, the company needs to maintain integrations with popular accounting software packages such as QuickBooks, Netsuite, Sage Intacct and Xero. Integration occurs through APIs made available by these software vendors. If any of these accounting software providers change the features in their APIs or discontinue support this would negatively impact Bill.com's business. In addition, Bill.com continues to evolve its platform to changes in popular accounting software packages and if that is not possible, the company's business would be negatively impacted.

DOCS

- Doximity competes in a market with low relative switching costs. While the company currently enjoys a significant competitive advantage given the high number of physicians on its platform, physicians could switch platforms at relatively little cost.

- A declining number of pharmaceuticals under patent could reduce revenue opportunity for Doximity's core marketing solutions segment. The vast majority of the marketing spend by Pharmaceutical companies on the Doximity platform is related to newer drugs currently under patent protection. A reduction in the number of drugs under patent likely reduces the need to high levels of marketing spend to promote these newer care options.

HUBS

- Key risks to our investment thesis include increased competition, SMB challenges, high investor expectations, and the possibility of profitability supplanting growth as an emphasis in investors' minds.

EMST

- A weakening macro environment could delay or reduce IT spending, and payment processing is cyclical in nature.

- Transactional element to business model provides less visibility into revenue streams.

- Movement into new verticals provides uncertainty regarding adoption rates, versatility of offering.

- Fragmented competitive landscape in which EngageSmart competes with both mature companies and start-ups. Competitive pressures could slow future top line growth.

FIVN

- Key risks to our investment thesis include 1) Macroeconomic deterioration, which could impact sales cycles, close rates, and retention; 2) well capitalized/aggressive competition, which could impact COA and pricing; 3) unknown adoption rates of new products including Social and Mobile customer care; and 4) volatility of SaaS valuations, which could drive multiple compression even with no changes to underlying fundamentals and opportunity.

PAYC

- We note that risks to our thesis and price target for Paycom include, but are not limited to, 1) competition and aggressive pricing; 2) macroeconomic headwinds, which could impact sales cycles and close rates; 3) unknown adoption rates of new products; and 4) potential regulatory changes

PCTY

- We note that risks to our thesis and price target for Paylocity include, but are not limited to, 1) competition and aggressive pricing; 2) macroeconomic headwinds, which could impact sales cycles and close rates; 3) unknown adoption rates of new products; and 4) potential regulatory changes.

SMAR

- SMAR competes in a highly competitive Work Management/Corporate Productivity space, in which many competitors are larger and better capitalized than SMAR. These larger competitors may be able to exert pricing pressure on the industry that would adversely affect SMAR's profitability.

- Weakening macro could reduce or delay IT spending.

- Smartsheet's Software-as-a-Service delivery model yields low switching costs to customers. As a result, customer retention rates may come under pressure should SMAR lose its competitive positioning within its target market or experience operational issues.

VEEV

- Veeva competes in mature end-markets with larger, better-capitalized competitors. Larger competitors may be able to exert pricing pressure that could negatively impact the company's earnings potential.

- Veeva has a high reliance on the availability of Salesforce.com's force.com platform. Any material downtime to the uptime of this platform may cause customers to seek alternative solutions.

- Weakening macro could reduce or delay IT spending.

WDAY

- Current economic conditions have reduced corporate IT spending over the past 36 months. Should these conditions continue for an extended period or worsen, demand for WDAY's products will most likely be negatively affected.

- WDAY competes in a highly competitive Enterprise Software industry, with many competitors that are larger and better capitalized. These larger competitors may be able to exert pricing pressure on the industry that would adversely affect WDAY's profitability.

- Workday's Software-as-a-Service delivery model yields low switching costs to customers. As a result, customer retention rates may come under pressure should WDAY lose its competitive positioning within its target market or experience operational issues.

ANALYST CERTIFICATION

We, Scott Berg, Joshua Reilly and Ryan MacDonald hereby certify that the views expressed in this research report accurately reflect our personal views about the subject company (ies) and its (their) securities. We, also certify that we, have not been, and will not be receiving direct or indirect compensation in exchange for expressing the specific recommendation(s) in this report.

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Buy	80	32
Hold	19	5
Underperform	< 1	0
Rating Suspended	0	0
Restricted	0	0

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The subject companies Doximity, Inc. and EngageSmart, LLC currently are or during the 12-month period preceding the date of distribution of this research were a client of the Firm and received investment banking services.

The research analyst and research associate have received compensation based upon various factors, including quality of research, investor client feedback, and the Firm's overall revenues, which includes investment banking revenues for the following: Doximity, Inc., EngageSmart, LLC, Five 9, Inc., HubSpot, Inc., Netflix, Inc., Paycom Software, Inc., Paylocity Holding Corp., Smartsheet Inc, Veeva Systems Inc., Vertex Pharmaceuticals, Inc. and Workday, Inc.

The Firm, at the time of publication, makes a market in the subject companies Five 9, Inc., Netflix, Inc., Paylocity Holding Corp., Vertex Pharmaceuticals, Inc. and Workday, Inc. .

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